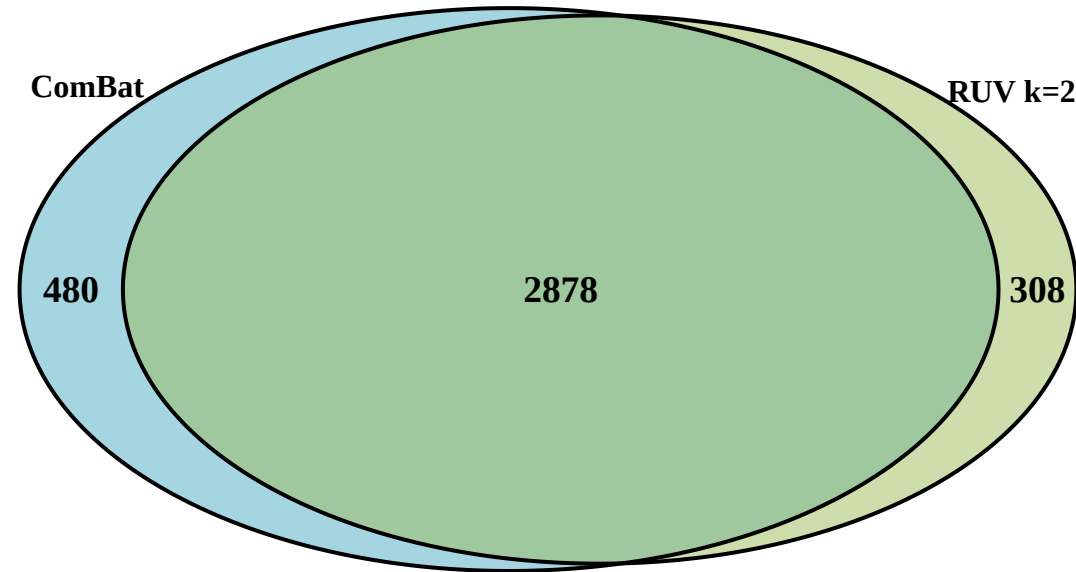
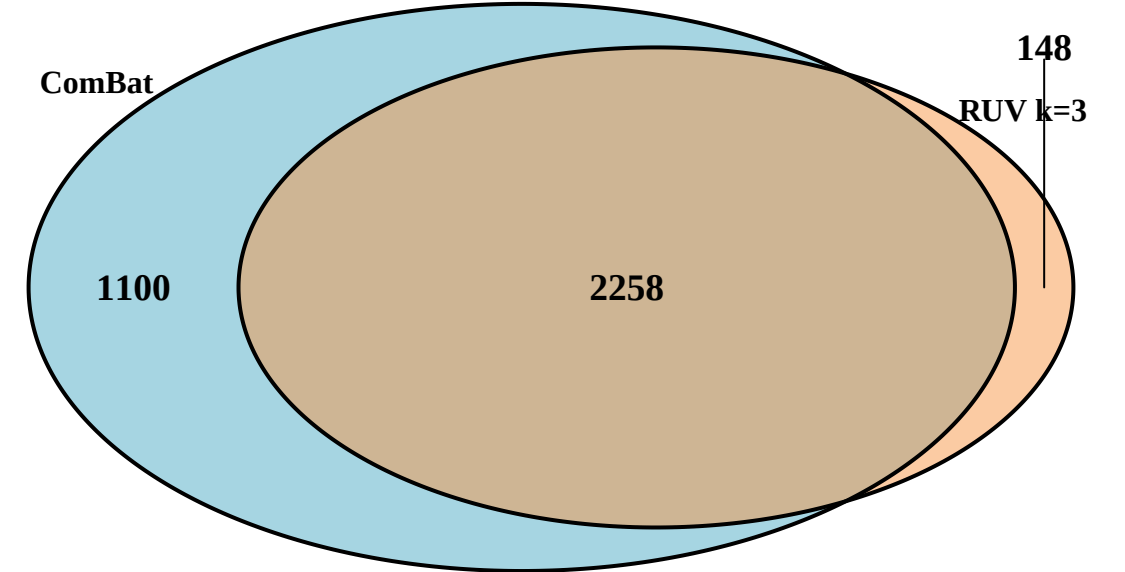


# 1st vs 2nd Trim – Balanced – DEG Venn Diagrams (FDR < 0.05)

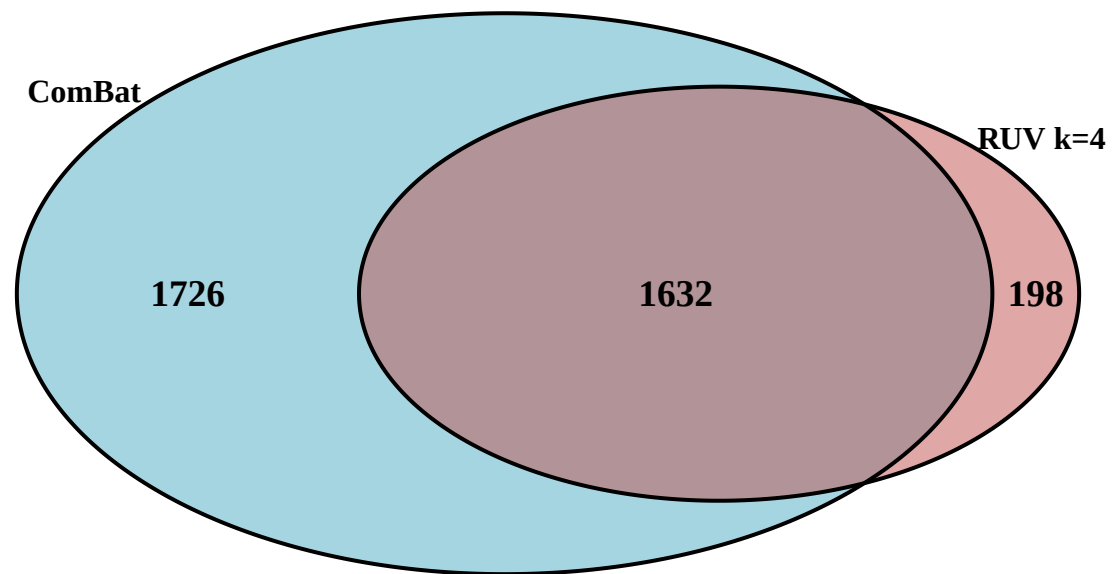
ComBat vs RUV k=2



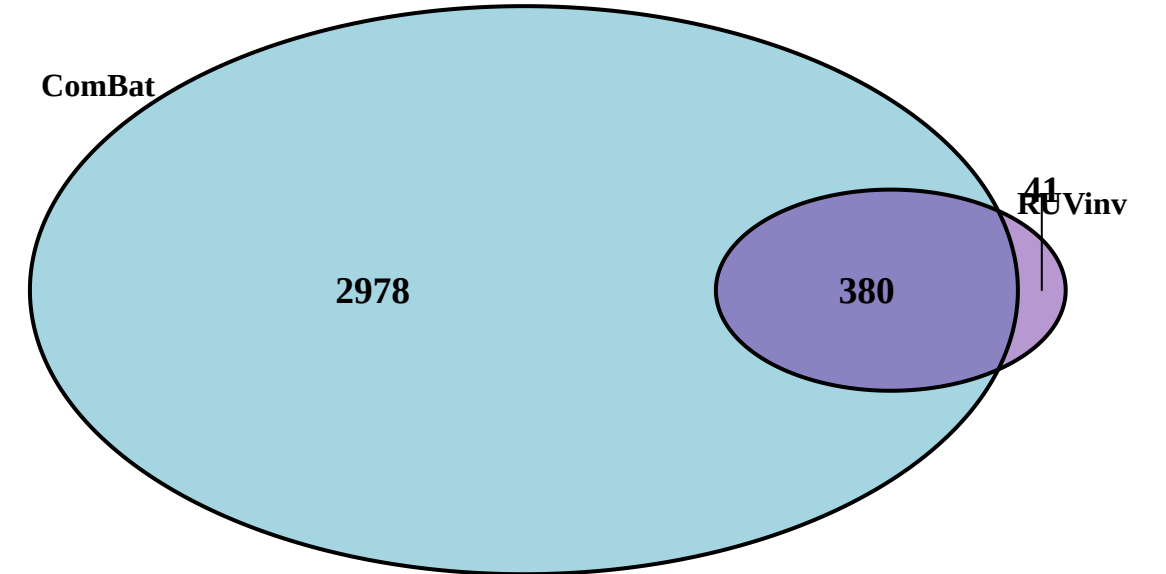
ComBat vs RUV k=3



ComBat vs RUV k=4

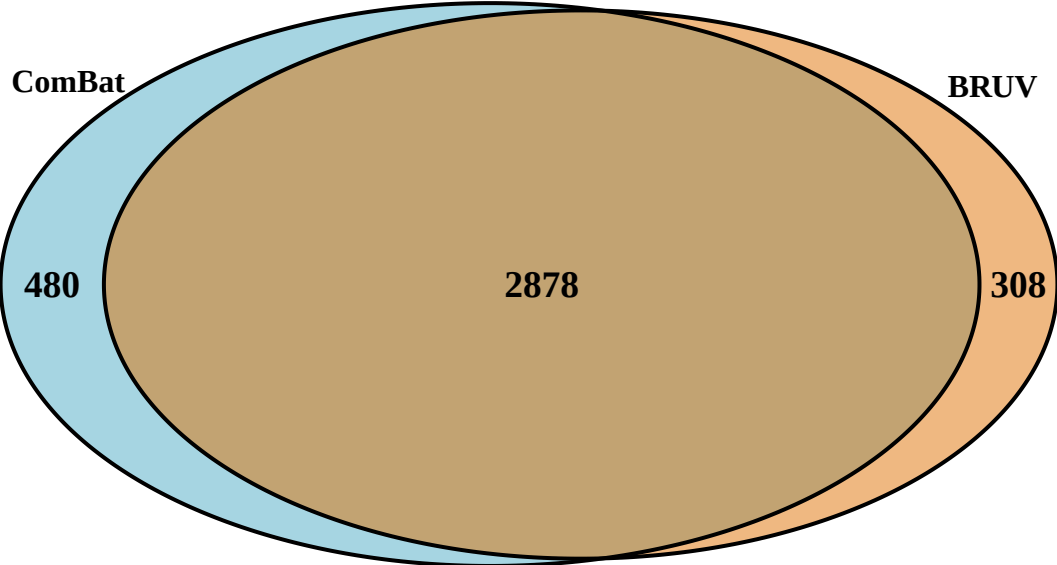


ComBat vs RUVinv

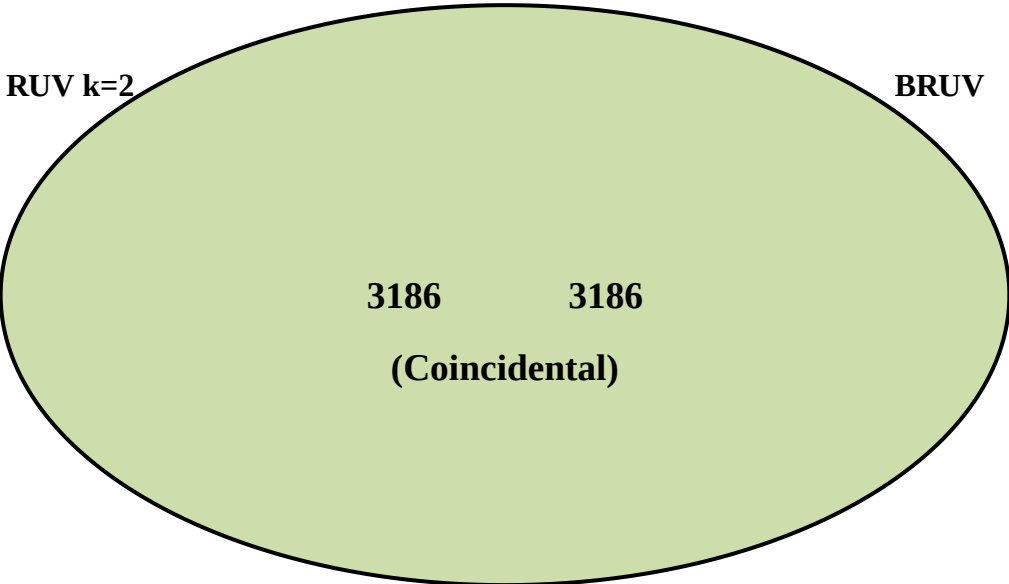


1st vs 2nd Trim – Balanced – BRUV Venn Diagrams (FDR < 0.05)

ComBat vs BRUV



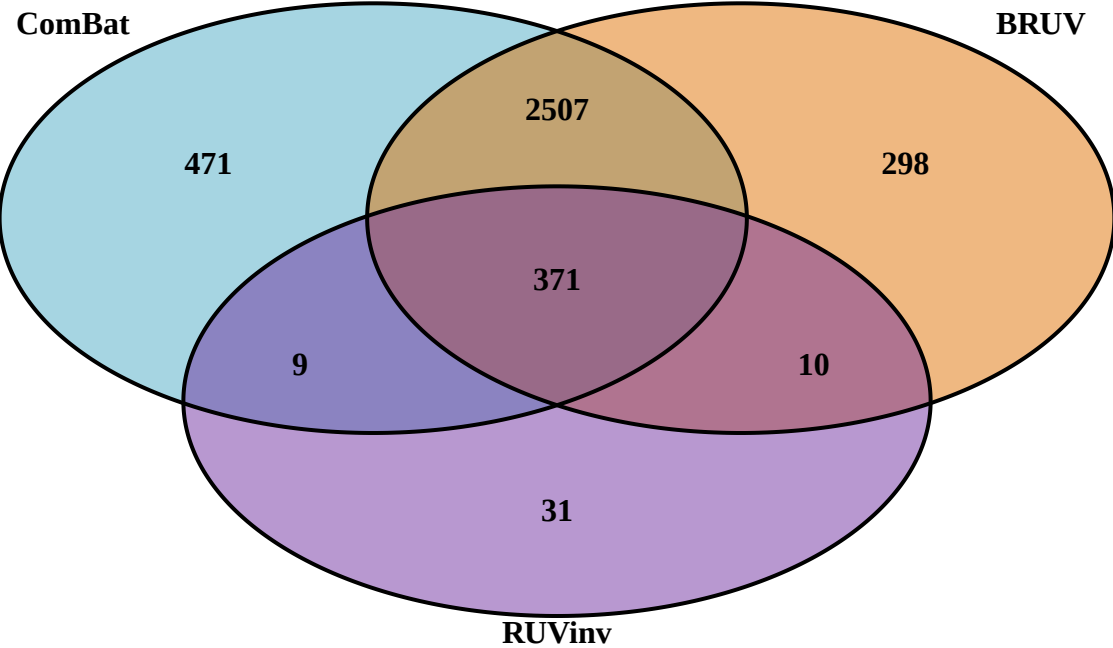
RUV k=2 vs BRUV



ComBat vs RUV k=2 vs BRUV

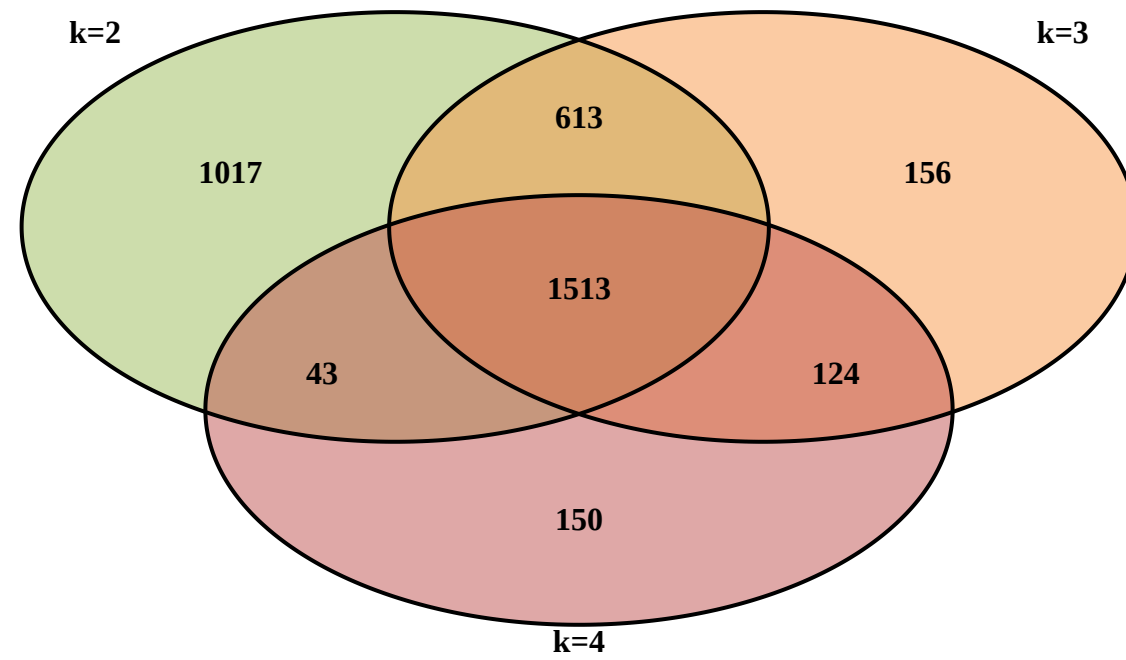


ComBat vs BRUV vs RUVinv

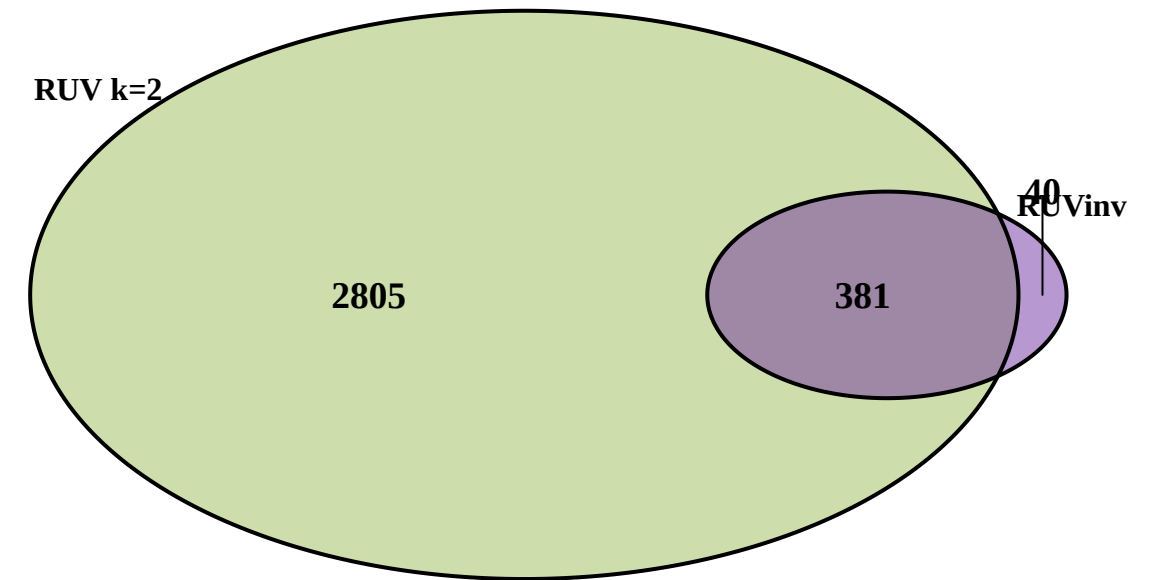


# 1st vs 2nd Trim – Balanced – RUV Variant Comparison (FDR < 0.05)

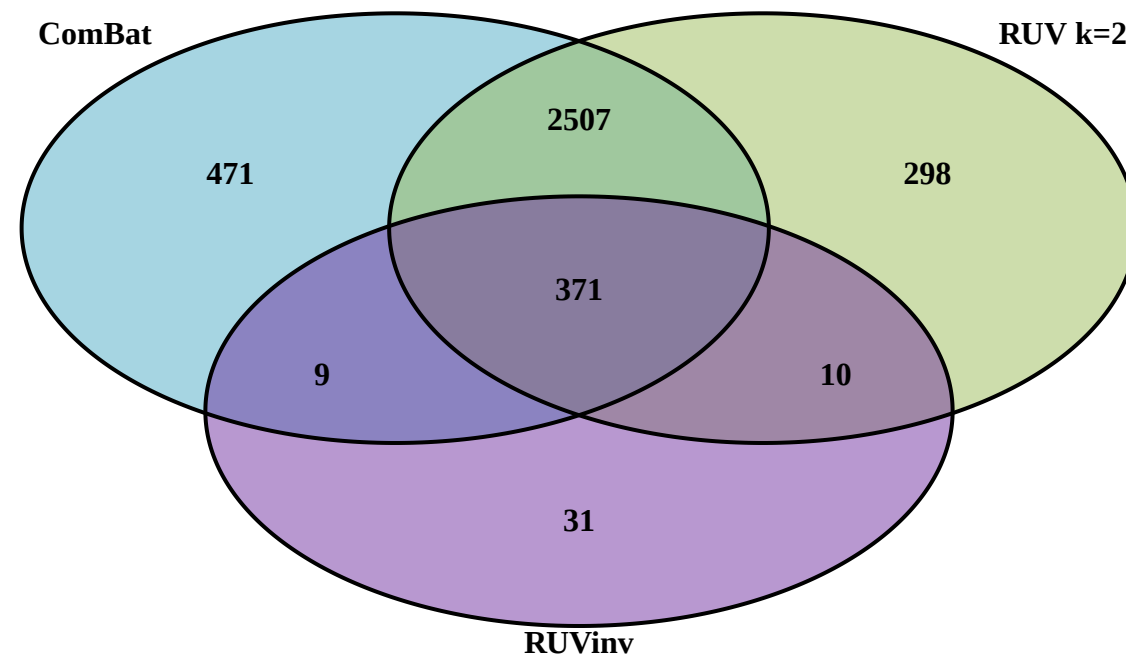
## RUV k=2 vs k=3 vs k=4



## RUV k=2 vs RUVinv

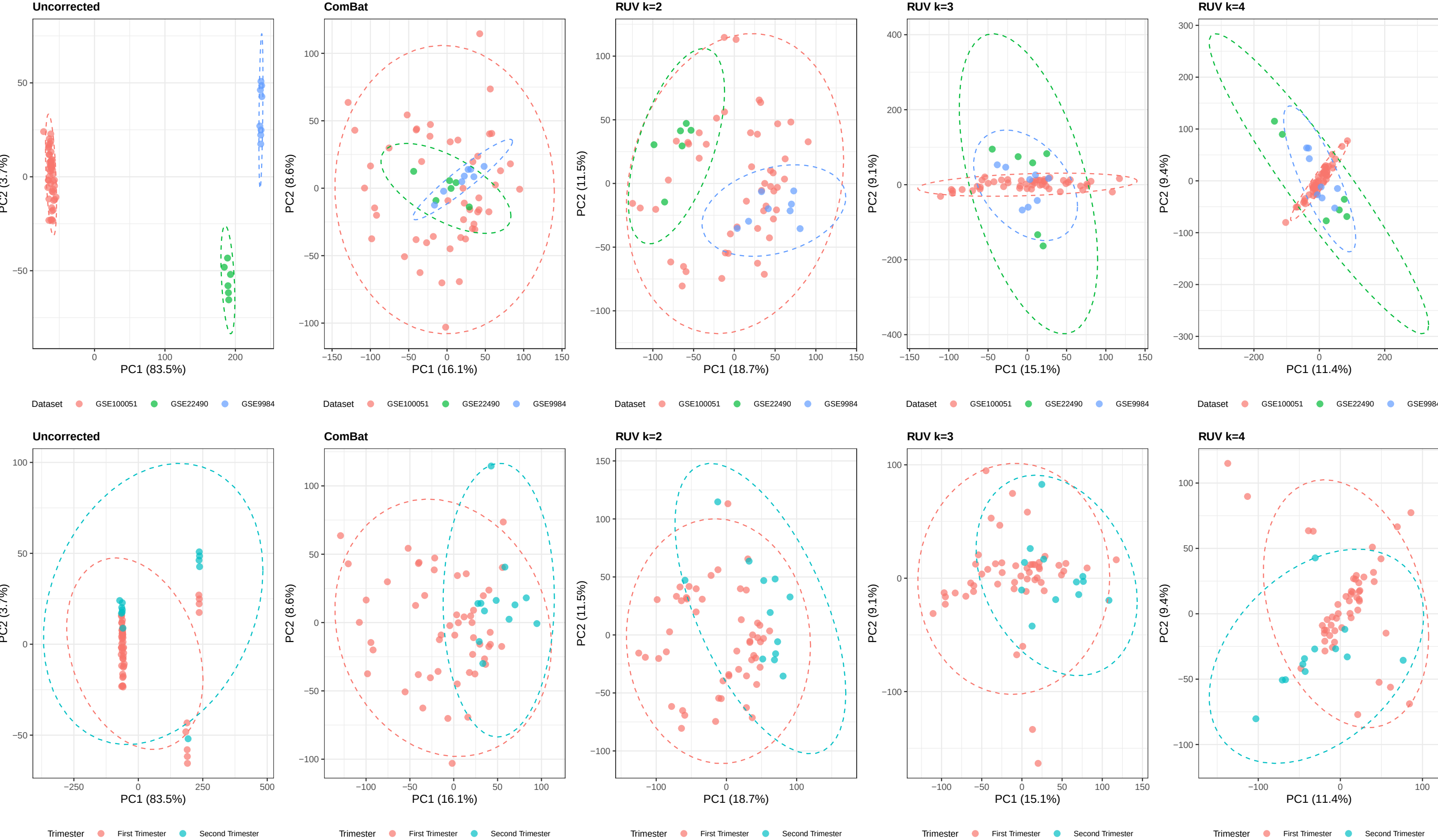


## ComBat vs RUV k=2 vs RUVinv



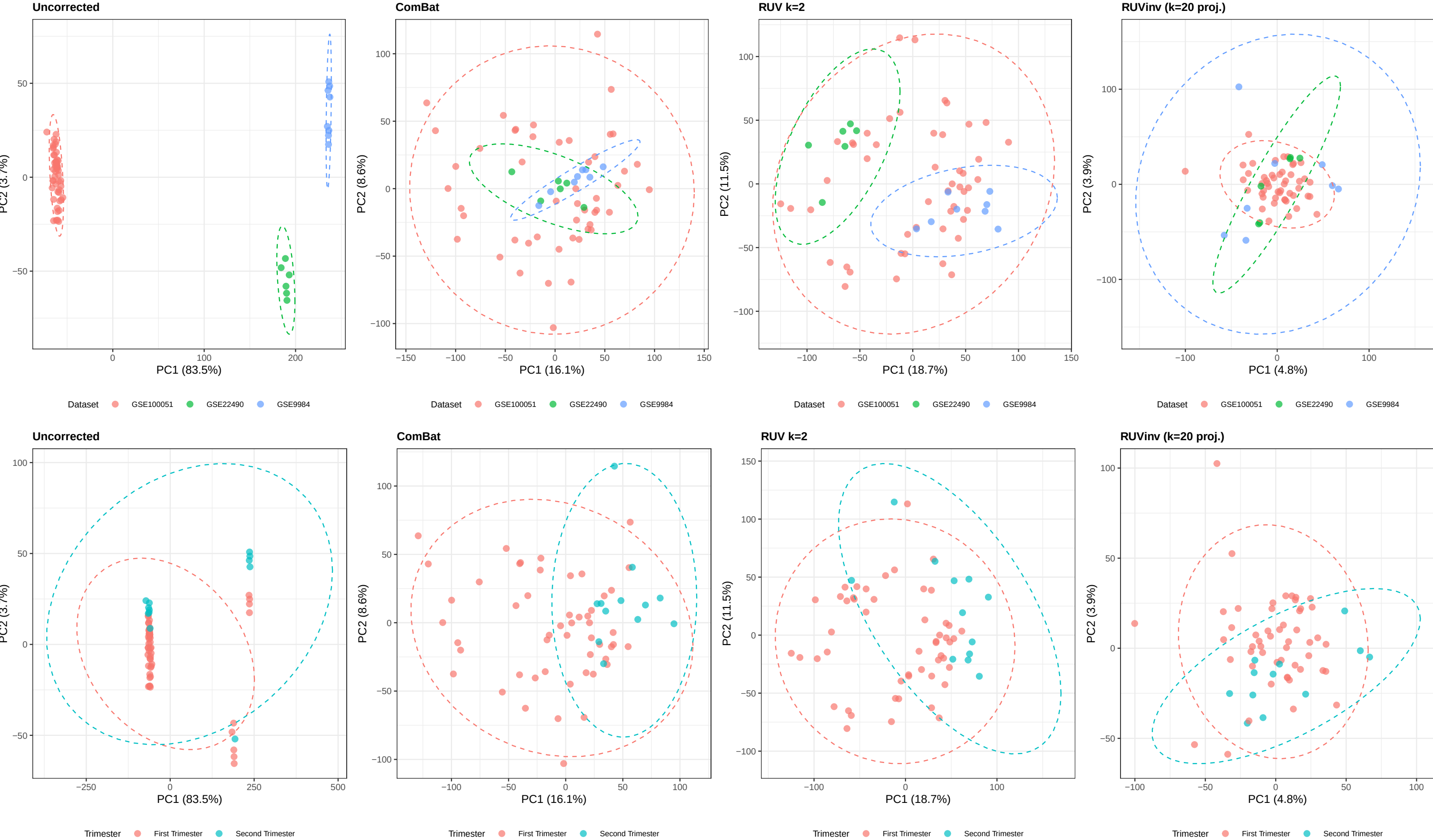
1st vs 2nd Trim – Balanced – PCA (k=2,3,4)

Top: colored by dataset (batch). Bottom: colored by trimester (biology).



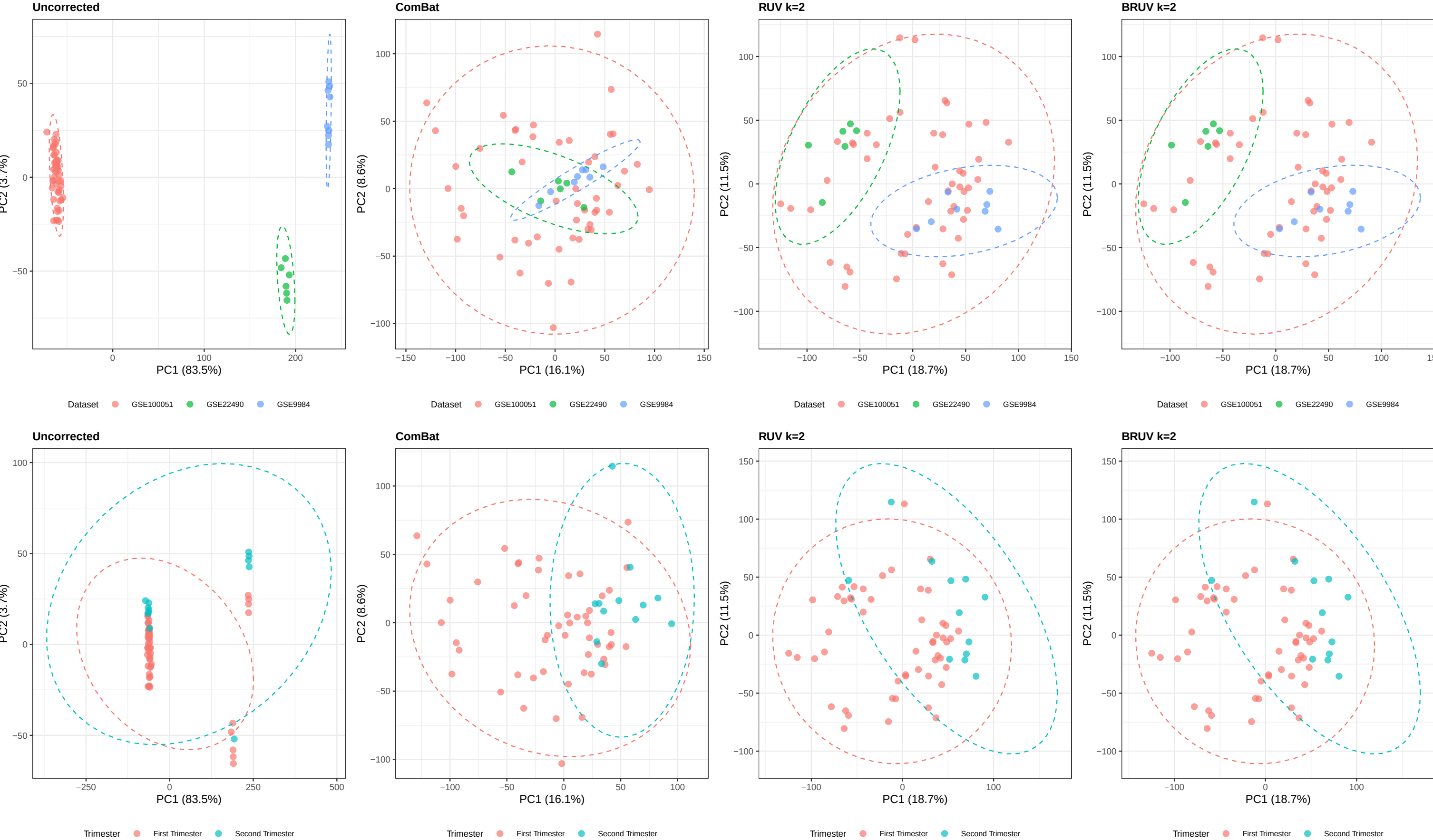
1st vs 2nd Trim – Balanced – PCA: Uncorrected / ComBat / RUV k=2 / RUVinv

Top: dataset. Bottom: trimester. RUVinv visualized by projecting out top-20 SVD components.



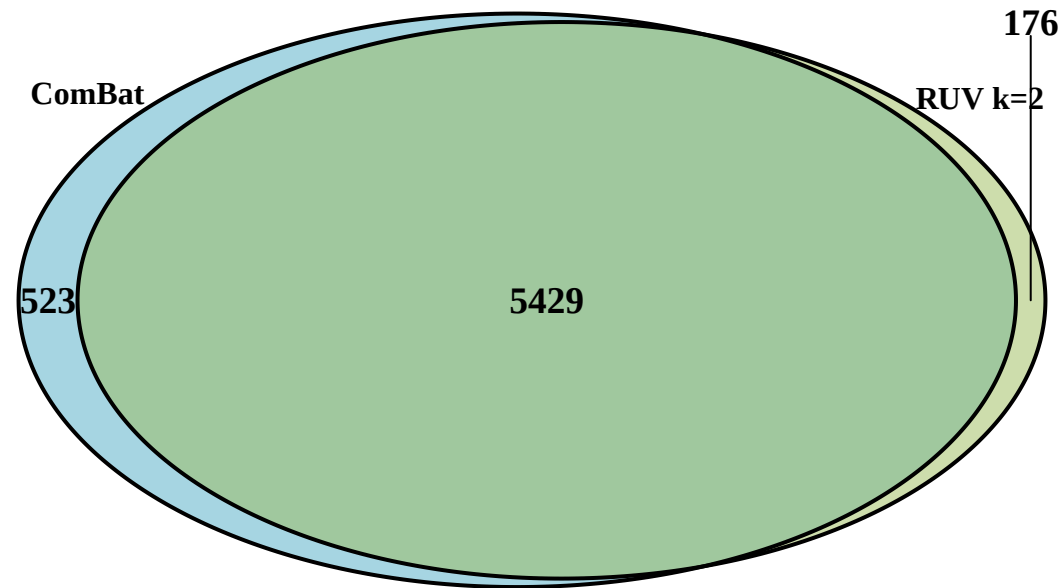
1st vs 2nd Trim – Balanced – PCA: Uncorrected / ComBat / RUV k=2 / BRUV

Top: dataset. Bottom: trimester. BRUV estimates W from balanced datasets only, projects onto all.



## 2nd Trim vs Term – Balanced – DEG Venn Diagrams (FDR < 0.05)

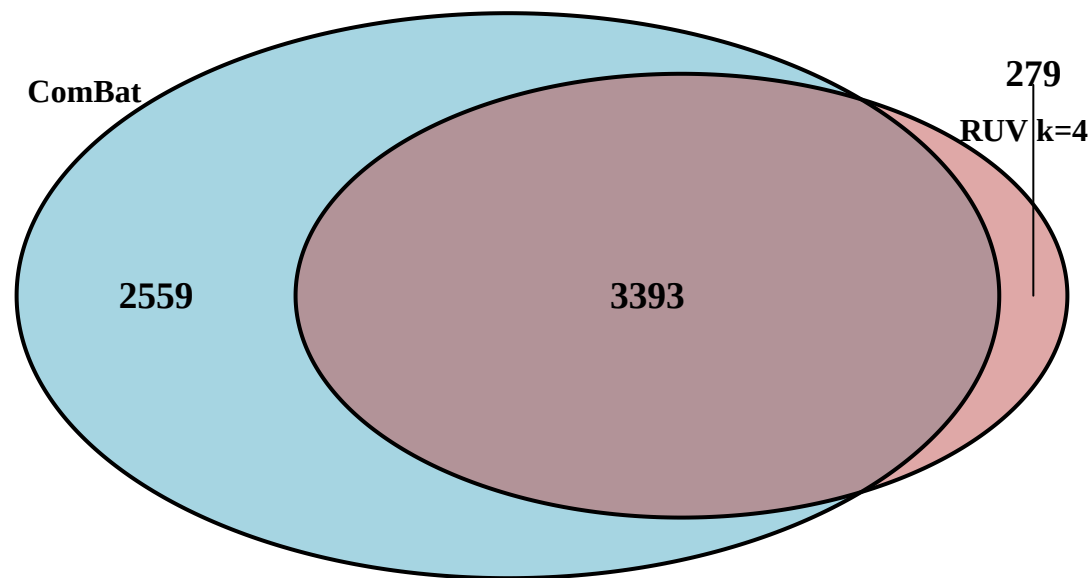
ComBat vs RUV k=2



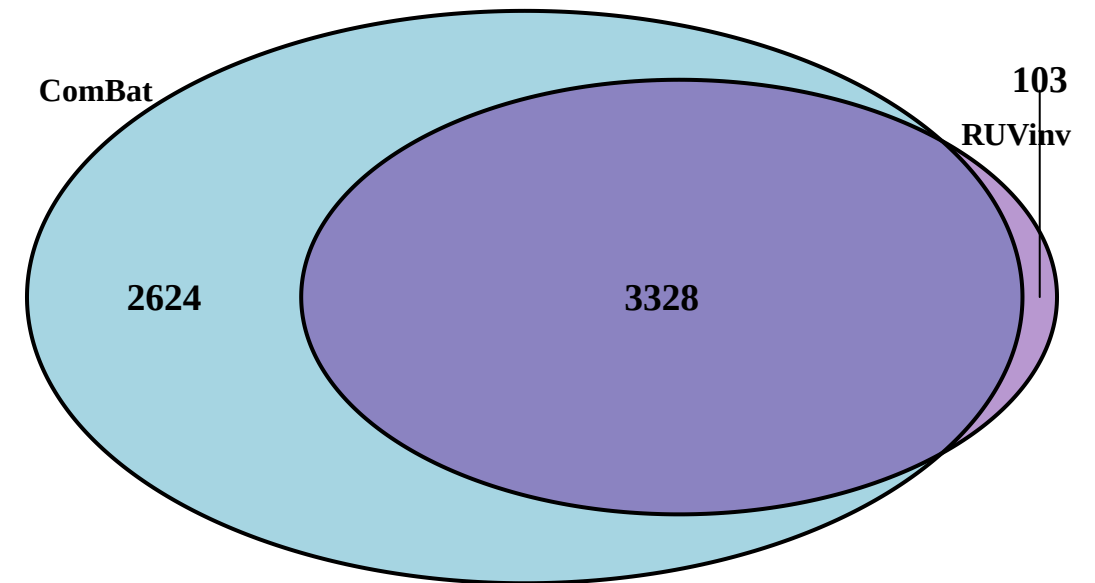
ComBat vs RUV k=3



ComBat vs RUV k=4



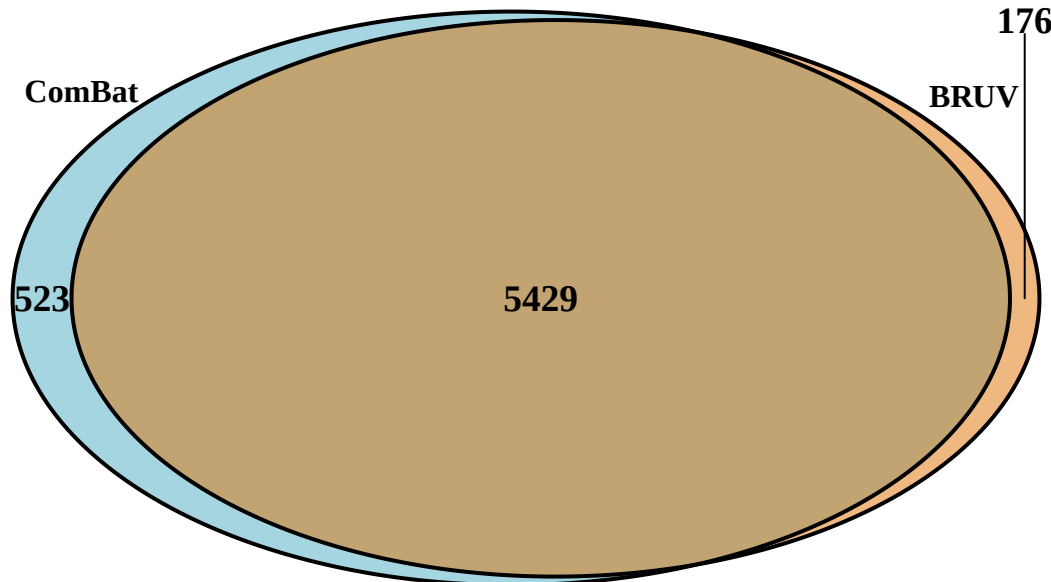
ComBat vs RUVinv



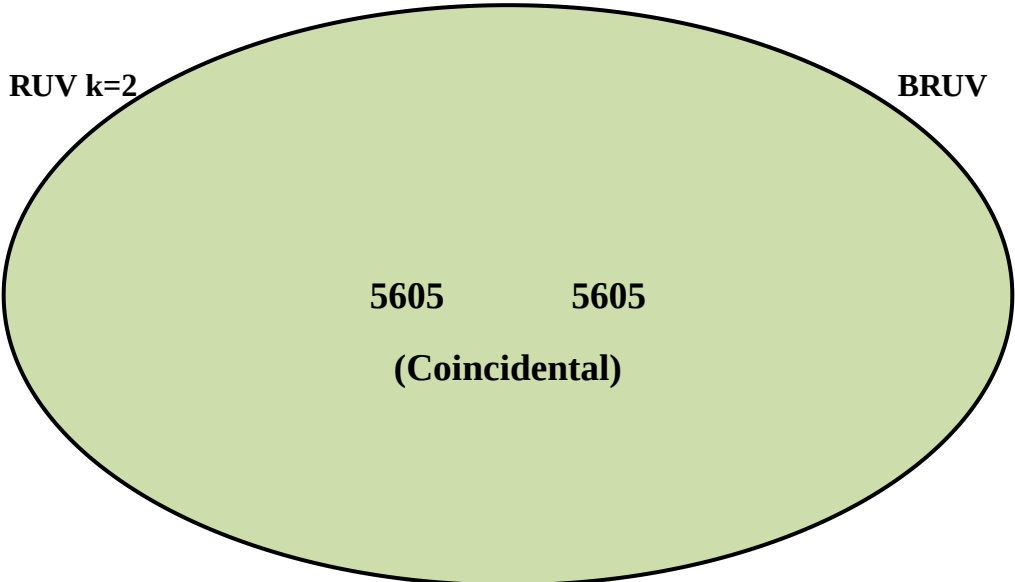


2nd Trim vs Term – Balanced – BRUV Venn Diagrams (FDR < 0.05)

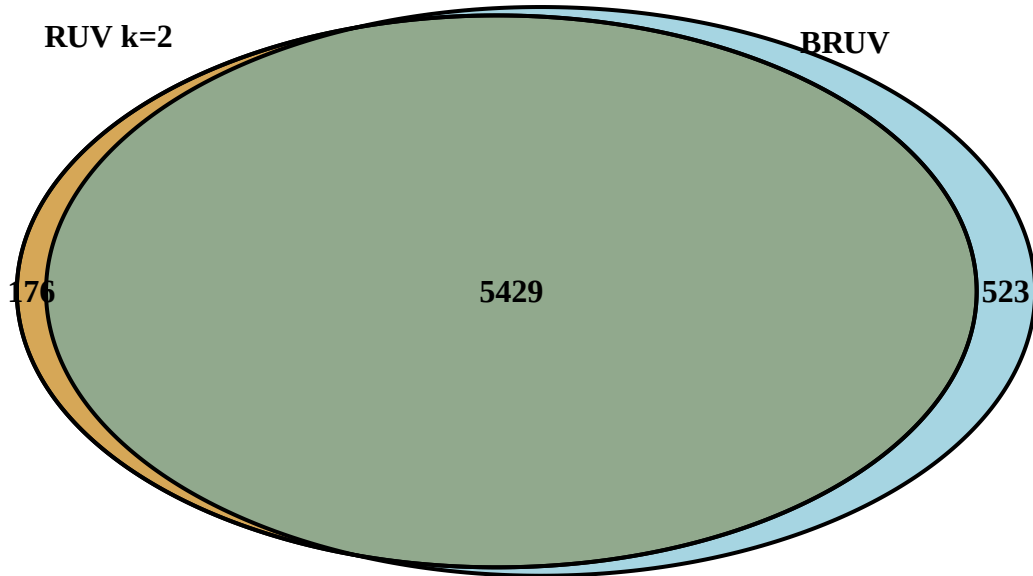
ComBat vs BRUV



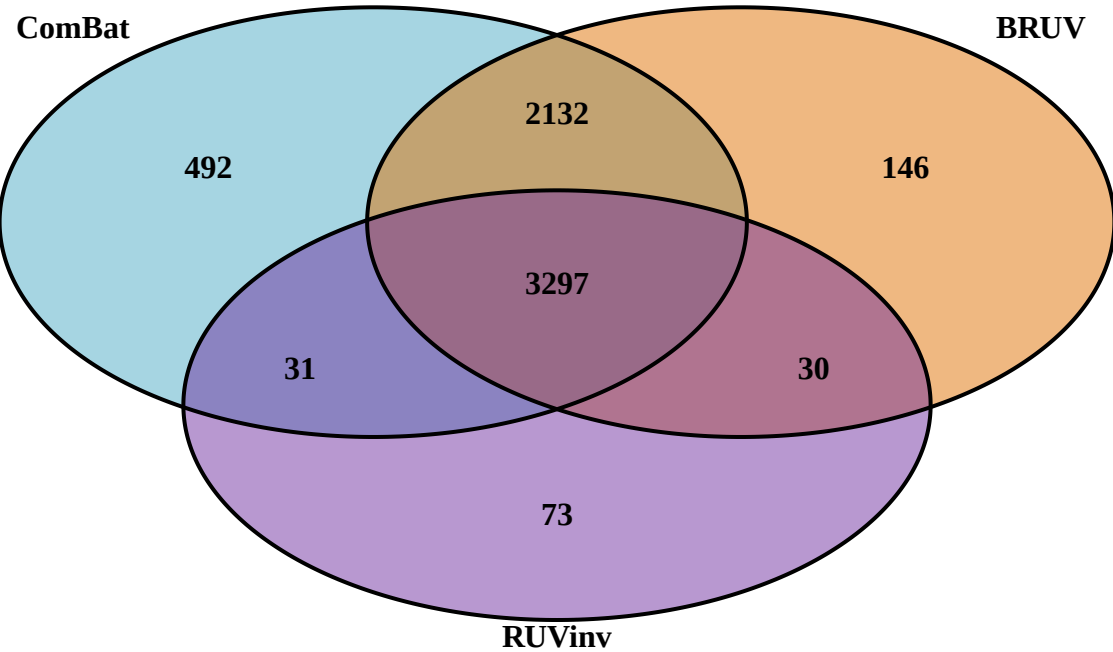
RUV k=2 vs BRUV



ComBat vs RUV k=2 vs BRUV



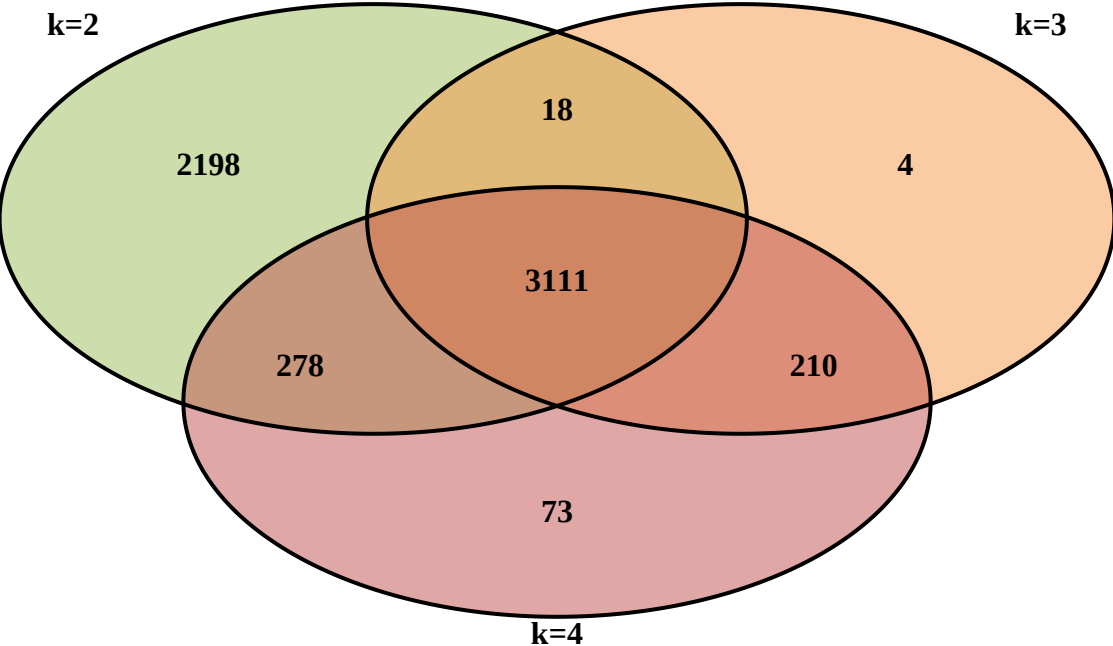
ComBat vs BRUV vs RUVinv



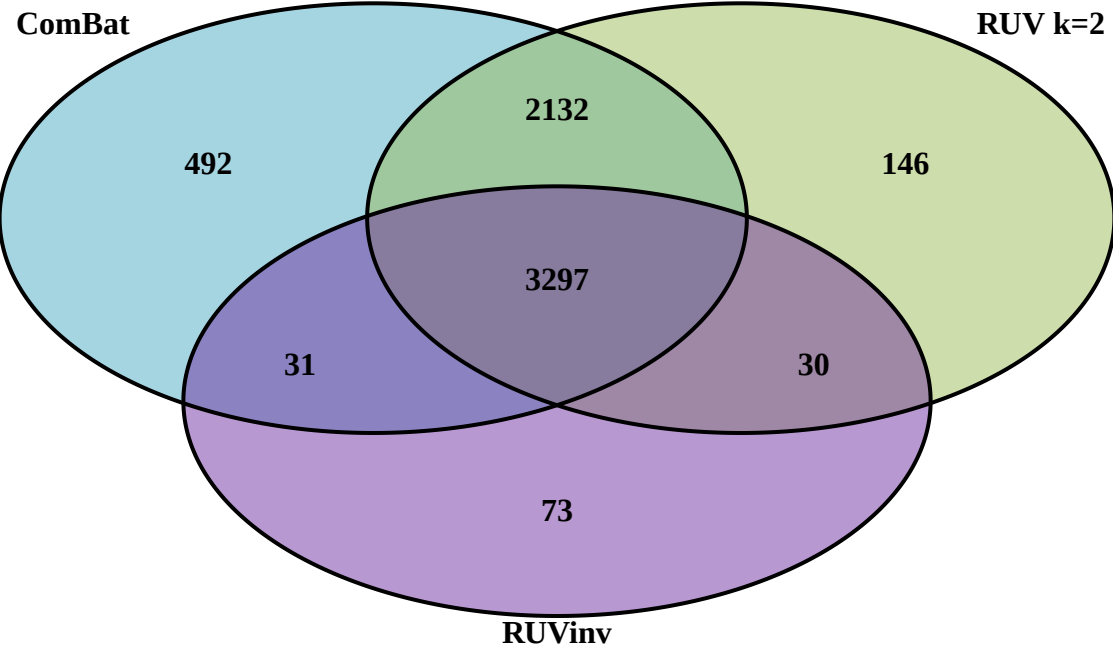


2nd Trim vs Term – Balanced – RUV Variant Comparison (FDR < 0.05)

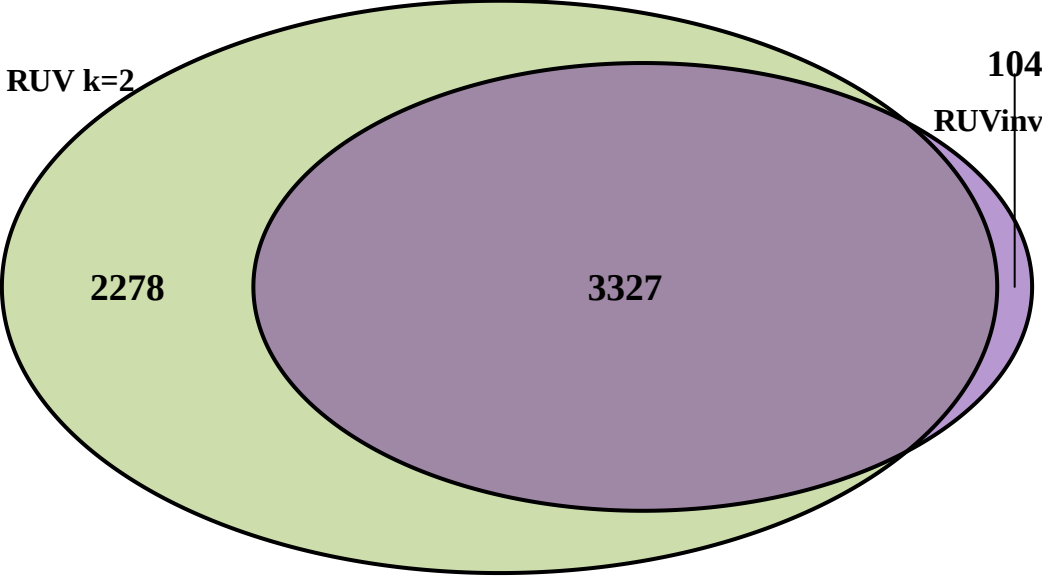
RUV k=2 vs k=3 vs k=4



ComBat vs RUV k=2 vs RUVinv

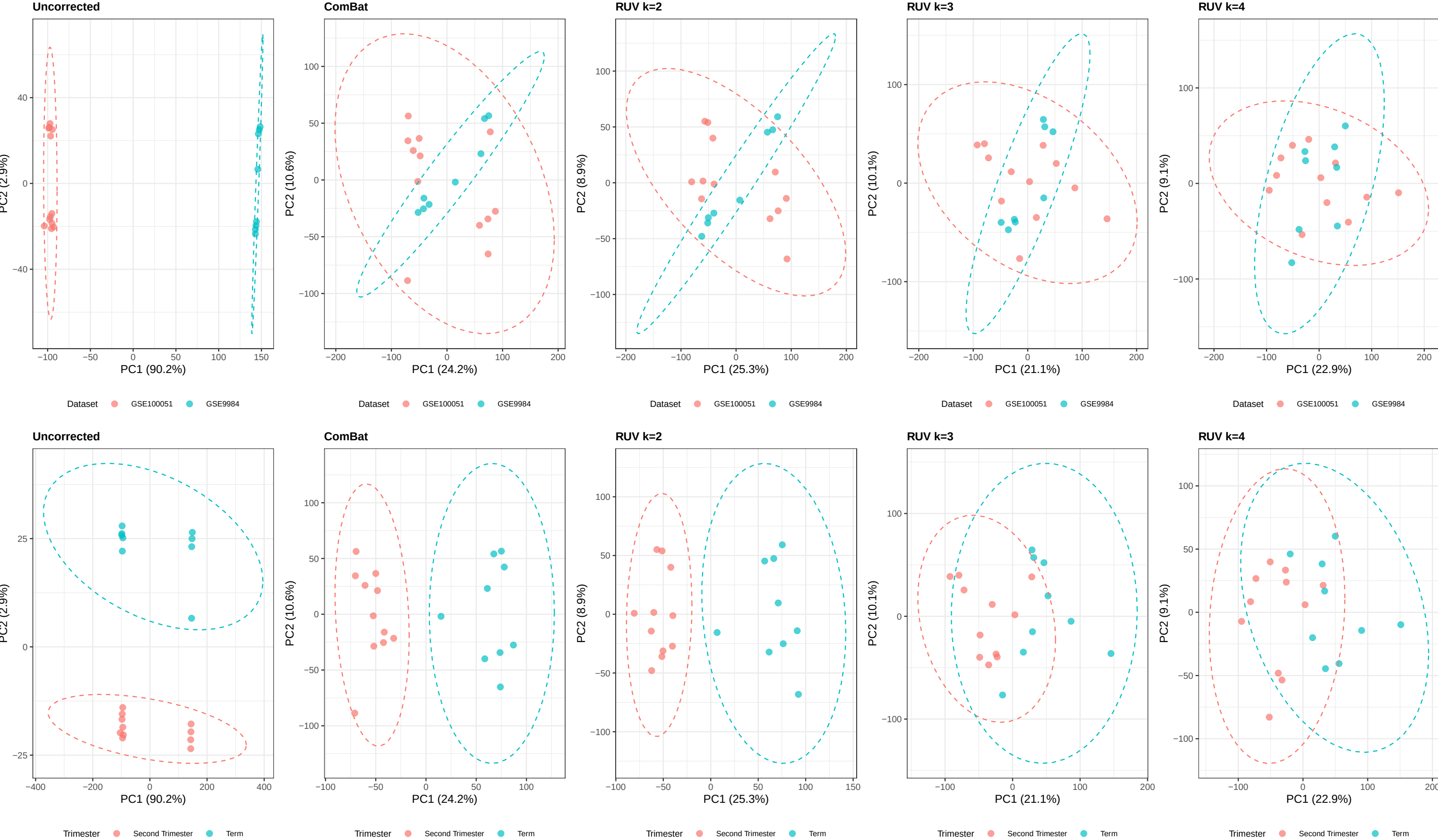


RUV k=2 vs RUVinv



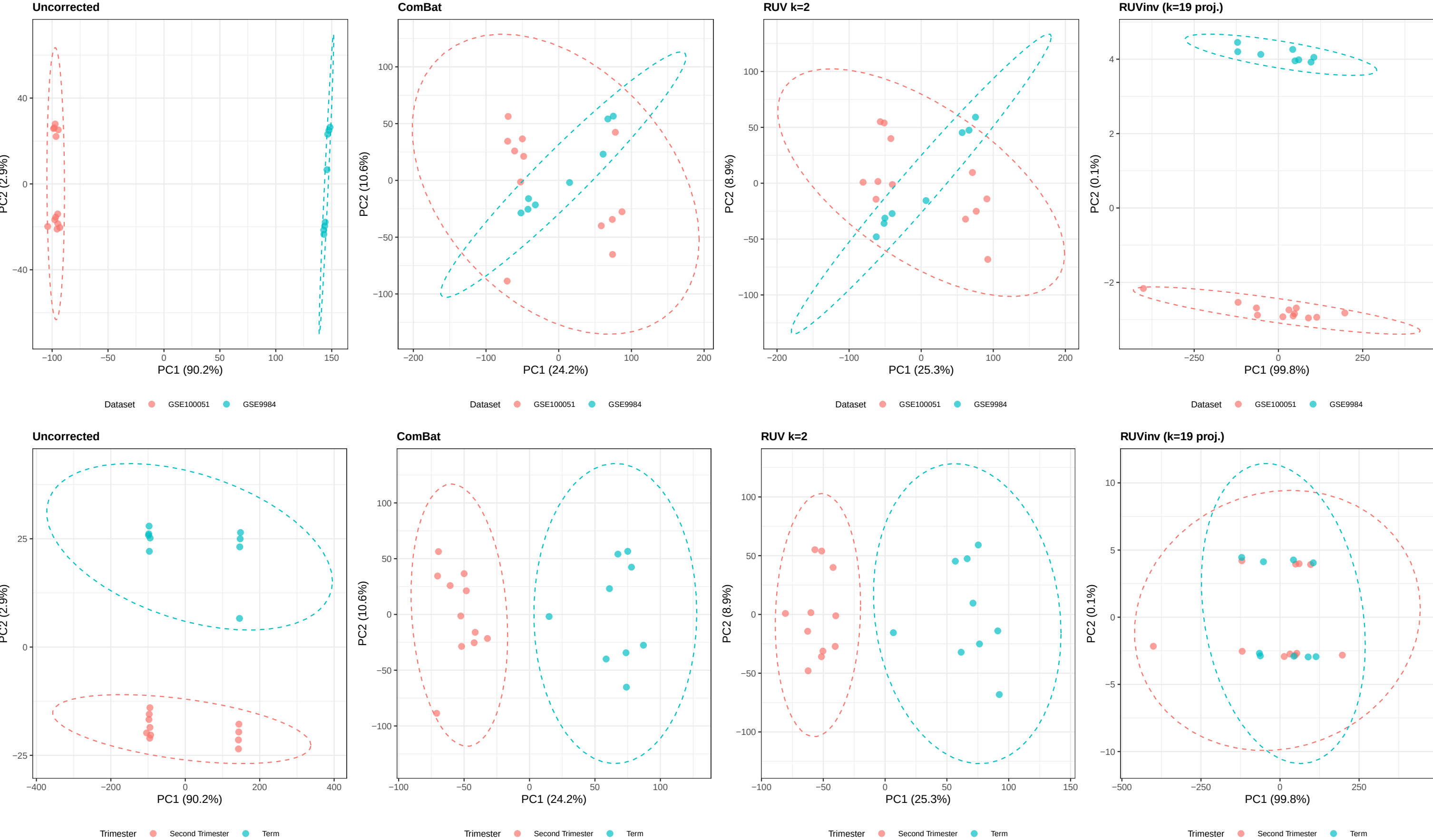
2nd Trim vs Term – Balanced – PCA (k=2,3,4)

Top: colored by dataset (batch). Bottom: colored by trimester (biology).



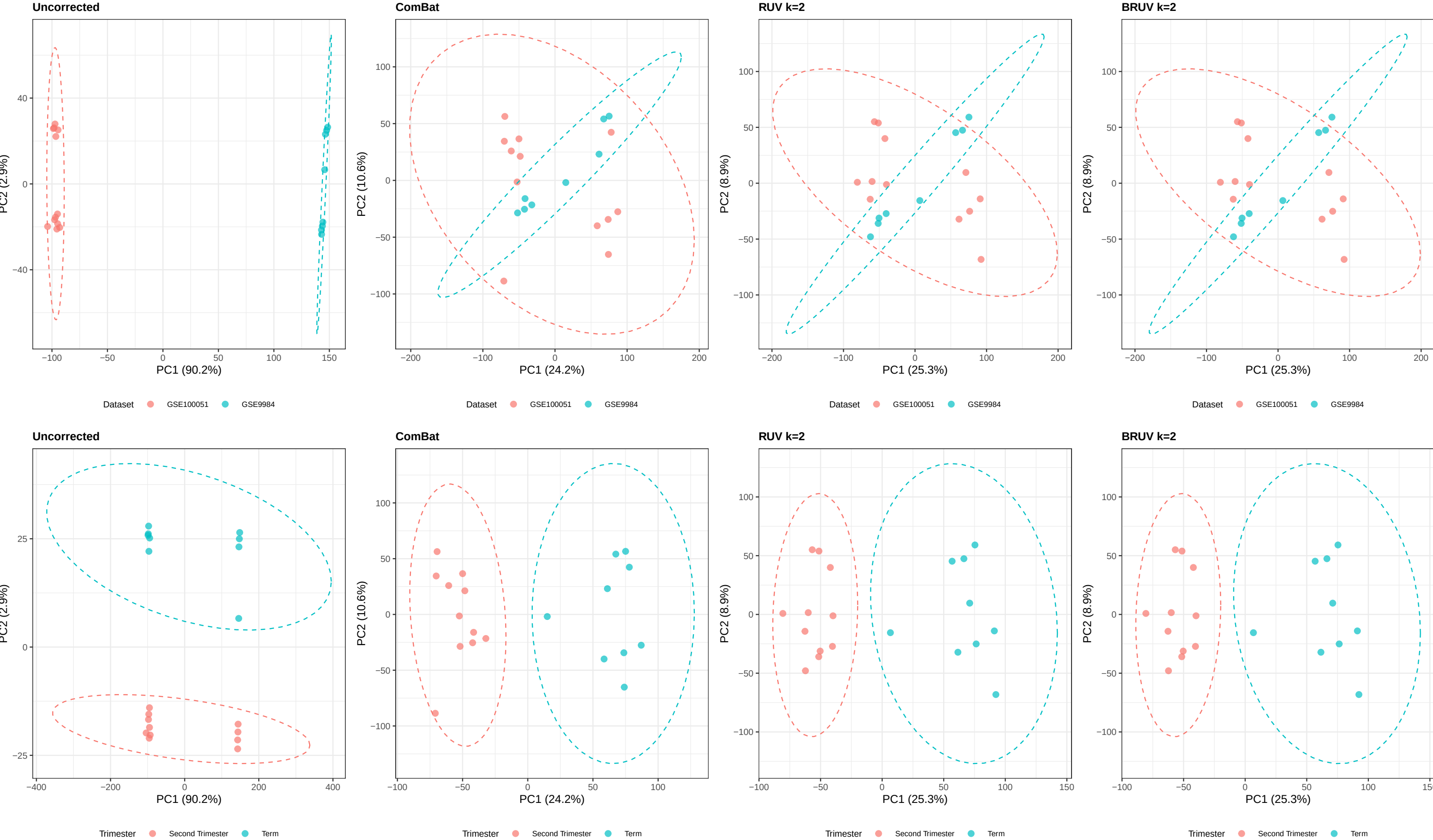
2nd Trim vs Term – Balanced – PCA: Uncorrected / ComBat / RUV k=2 / RUVinv

Top: dataset. Bottom: trimester. RUVinv visualized by projecting out top-20 SVD components.



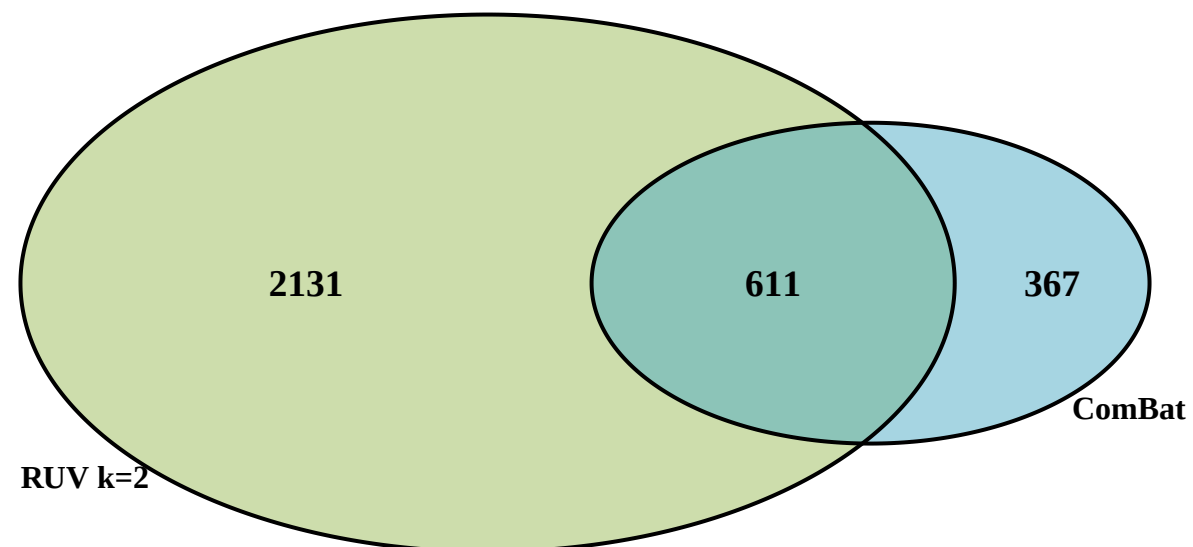
2nd Trim vs Term – Balanced – PCA: Uncorrected / ComBat / RUV k=2 / BRUV

Top: dataset. Bottom: trimester. BRUV estimates W from balanced datasets only, projects onto all.

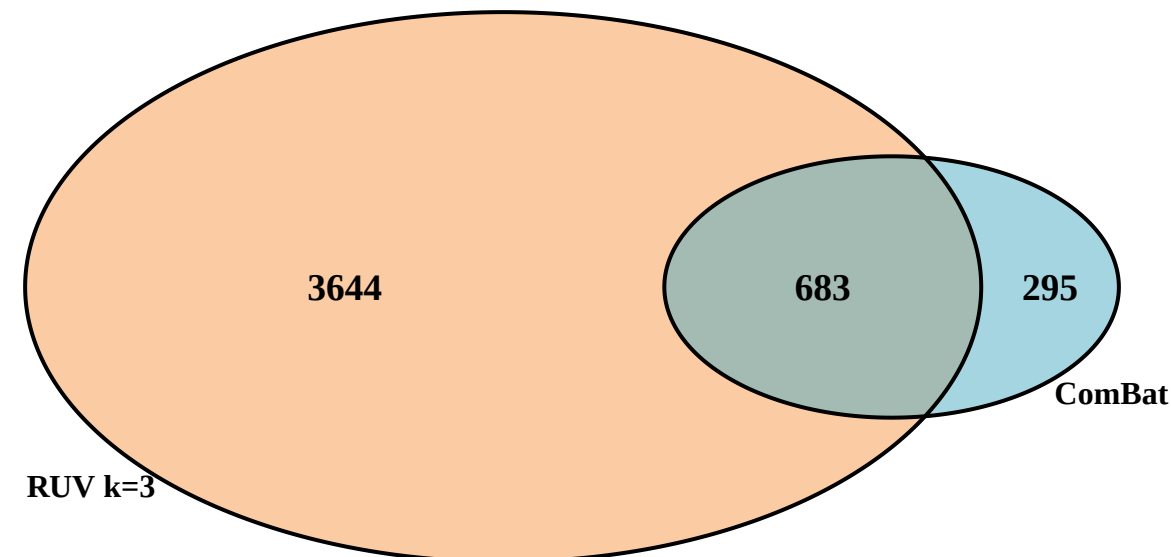


# 1st vs 2nd Trim – All Datasets – DEG Venn Diagrams (FDR < 0.05)

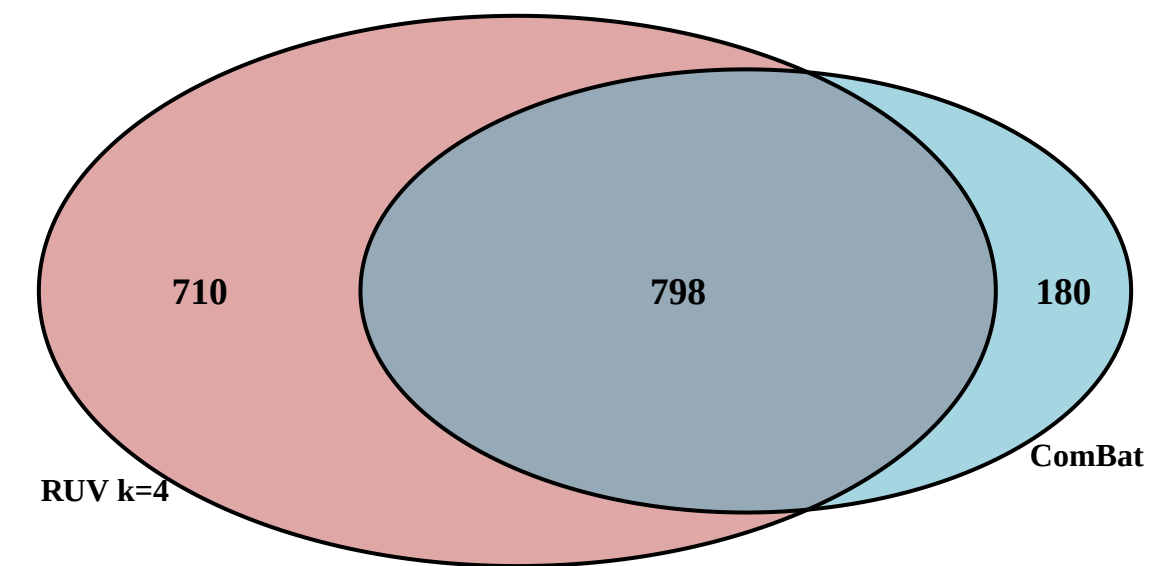
ComBat vs RUV k=2



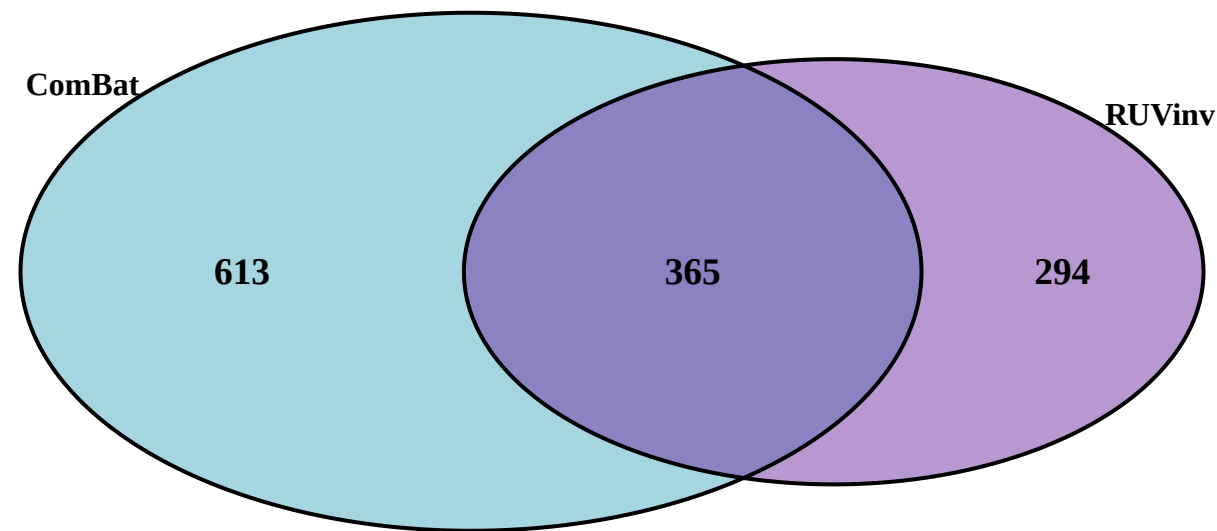
ComBat vs RUV k=3



ComBat vs RUV k=4

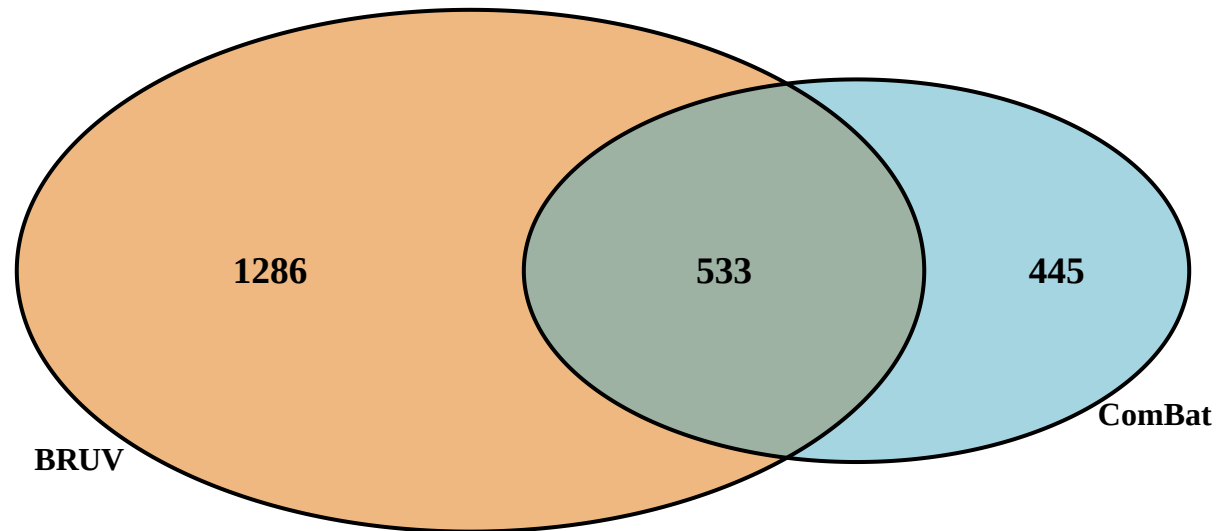


ComBat vs RUVinv

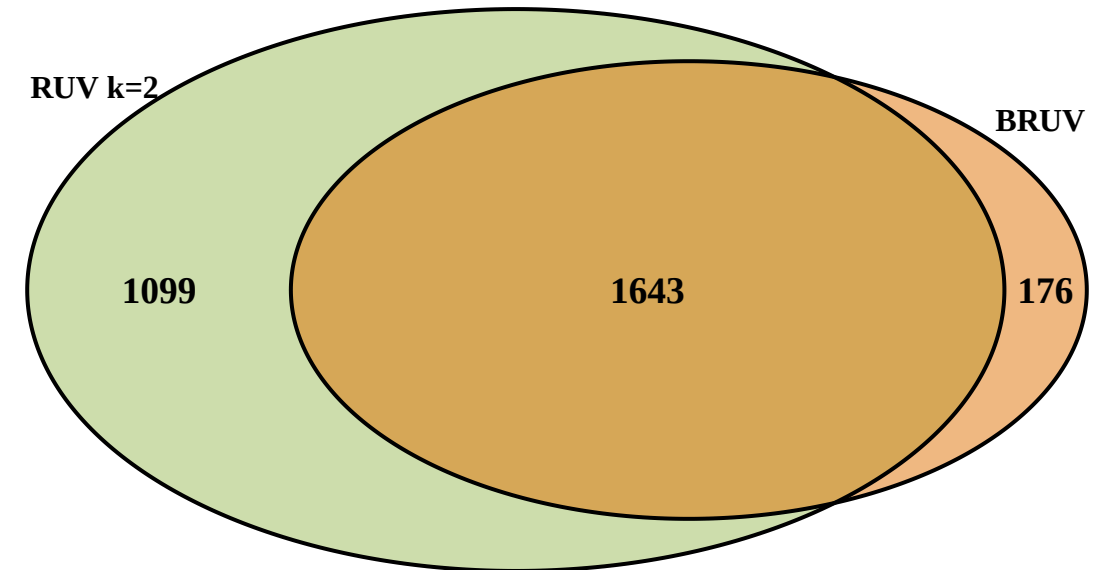


# 1st vs 2nd Trim – All Datasets – BRUV Venn Diagrams (FDR < 0.05)

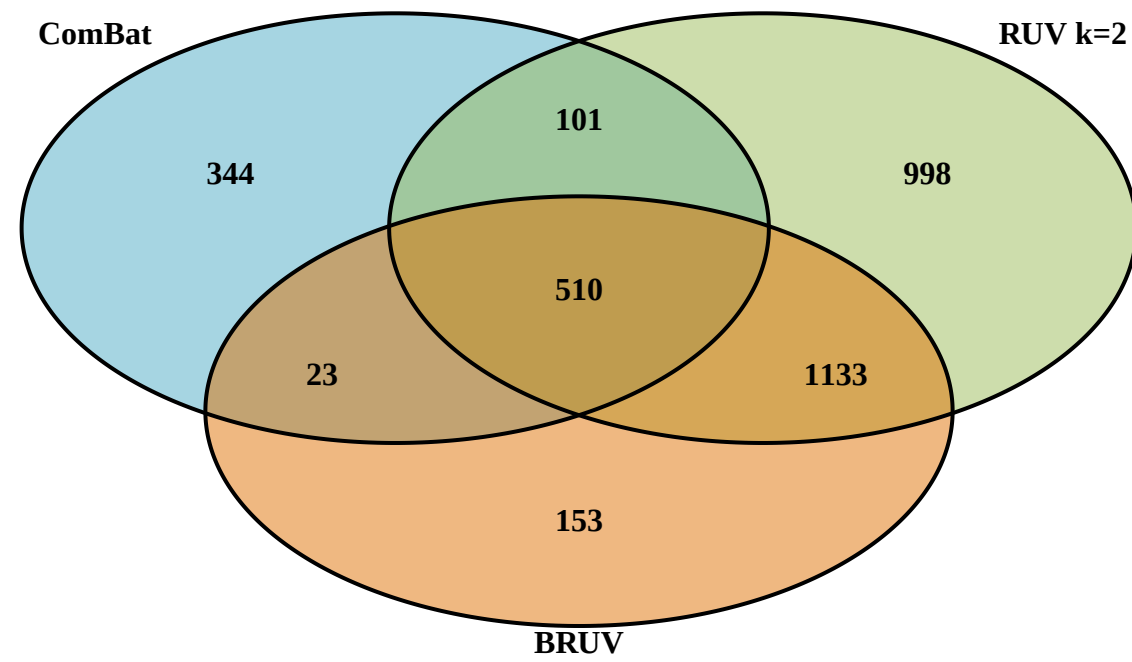
## ComBat vs BRUV



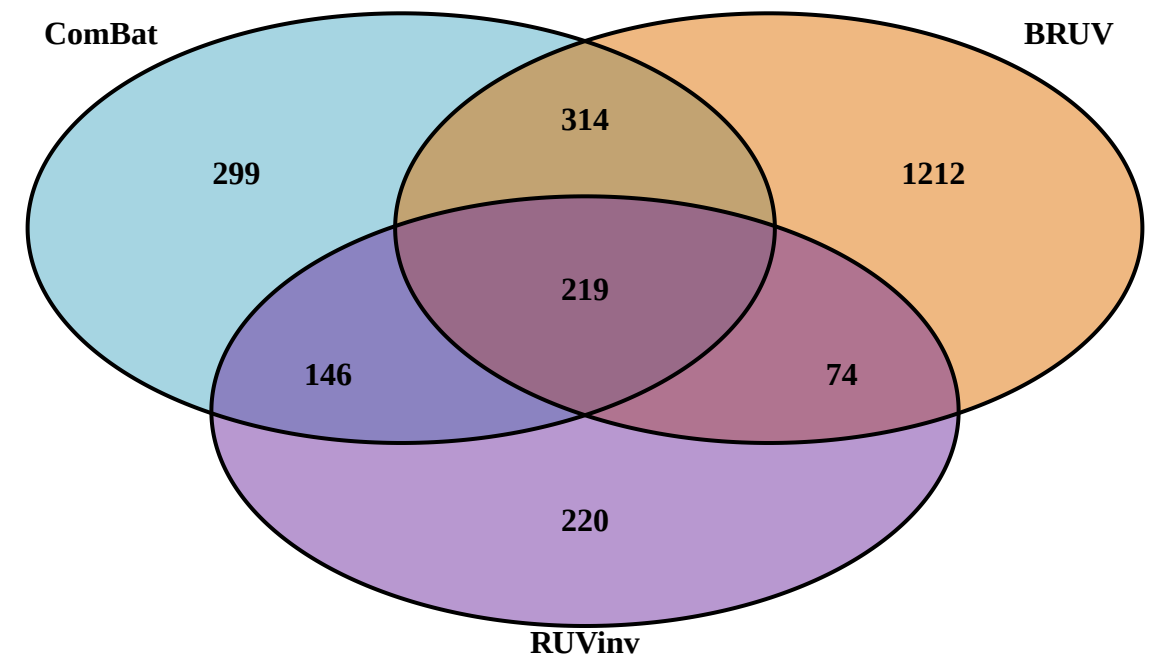
## RUV k=2 vs BRUV



## ComBat vs RUV k=2 vs BRUV

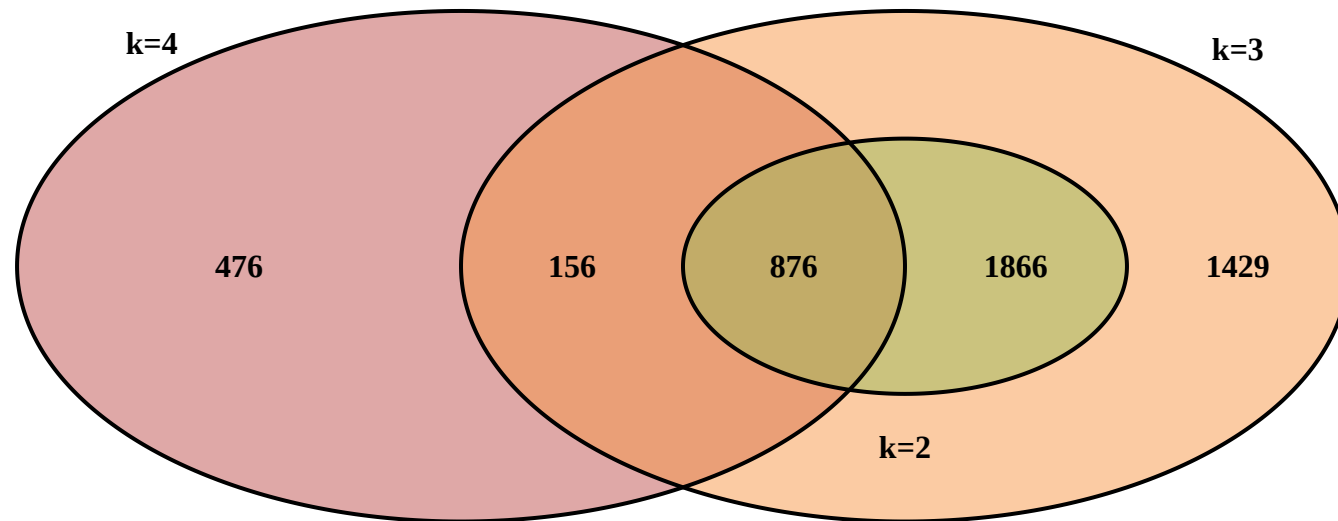


## ComBat vs BRUV vs RUVinv

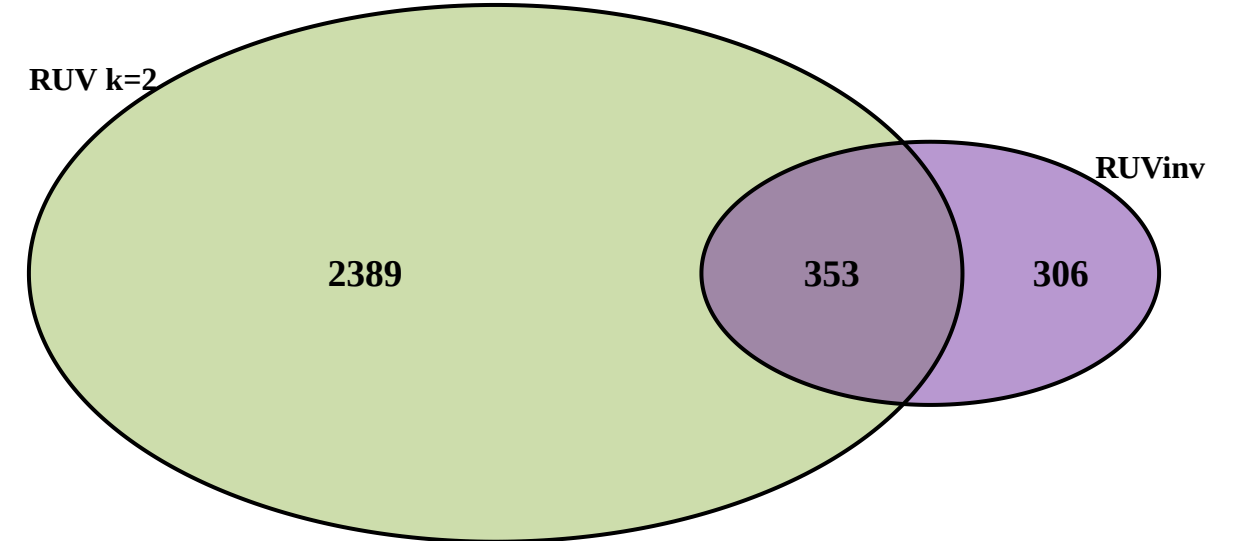


# 1st vs 2nd Trim – All Datasets – RUV Variant Comparison (FDR < 0.05)

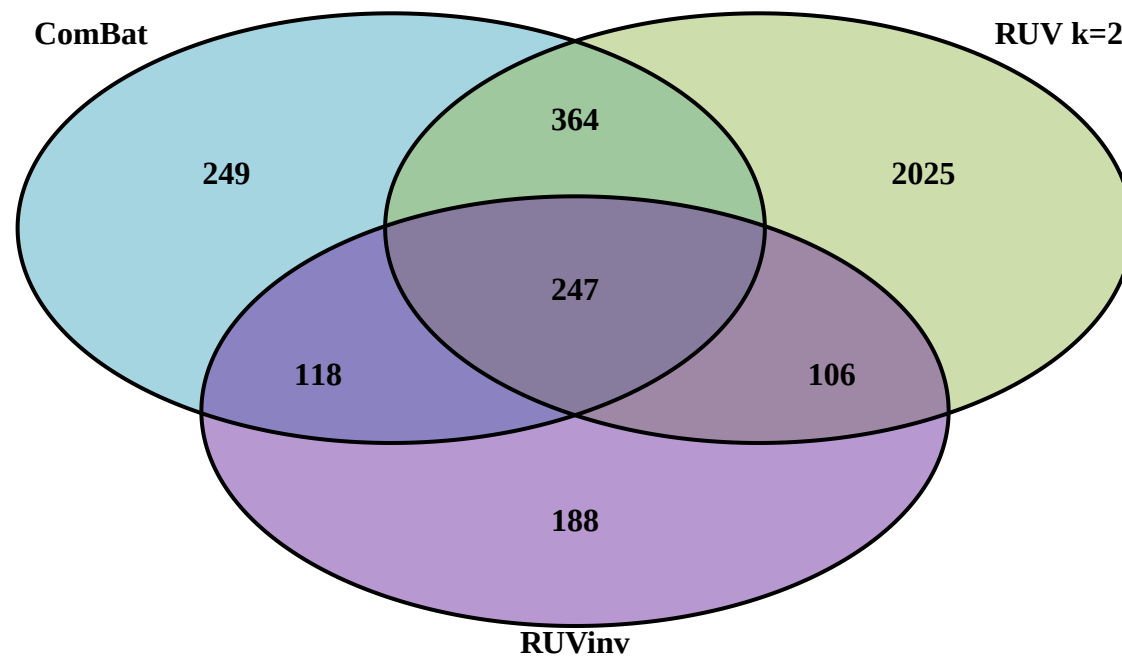
RUV k=2 vs k=3 vs k=4



RUV k=2 vs RUVinv



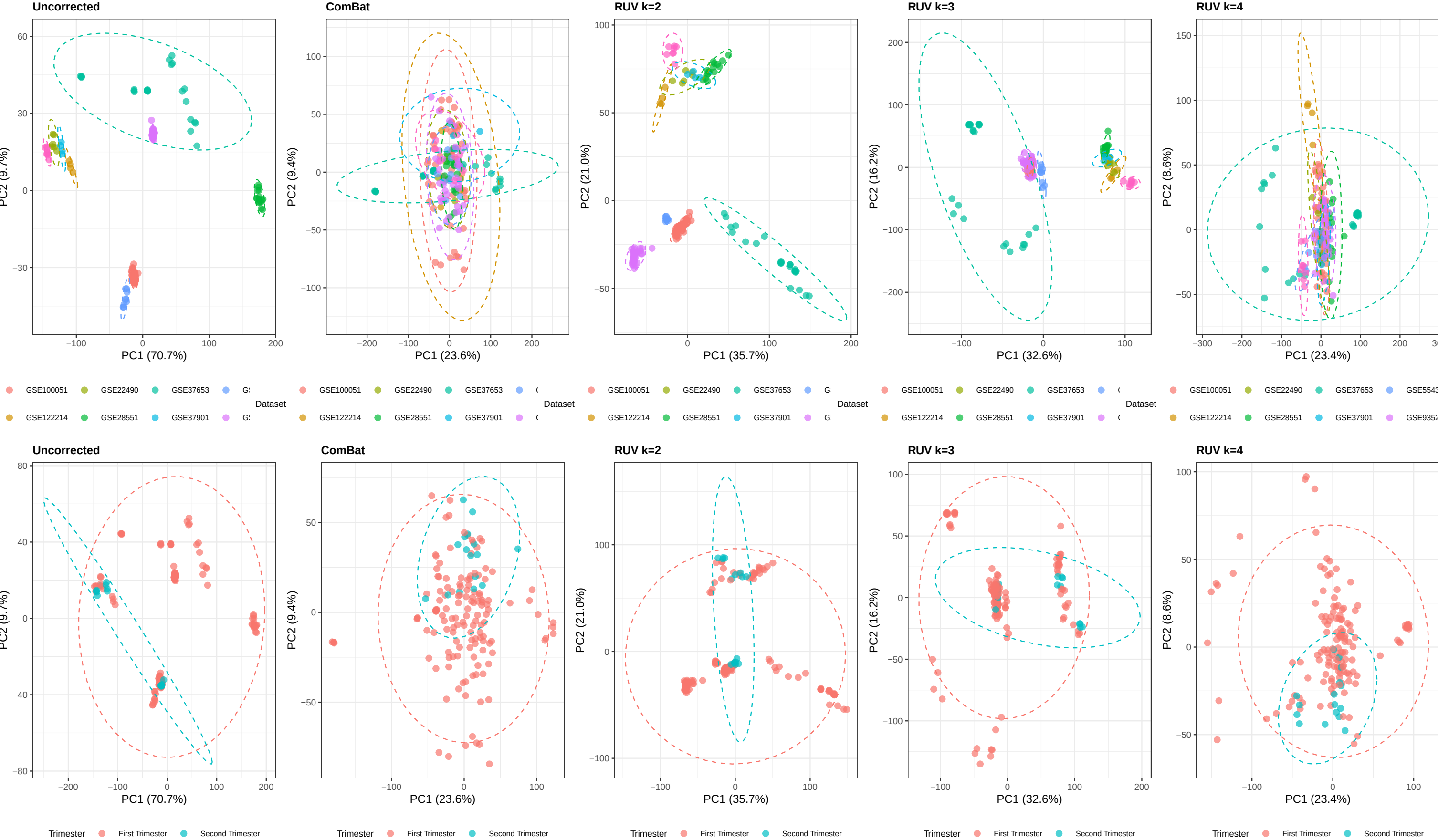
ComBat vs RUV k=2 vs RUVinv





1st vs 2nd Trim – All Datasets – PCA (k=2,3,4)

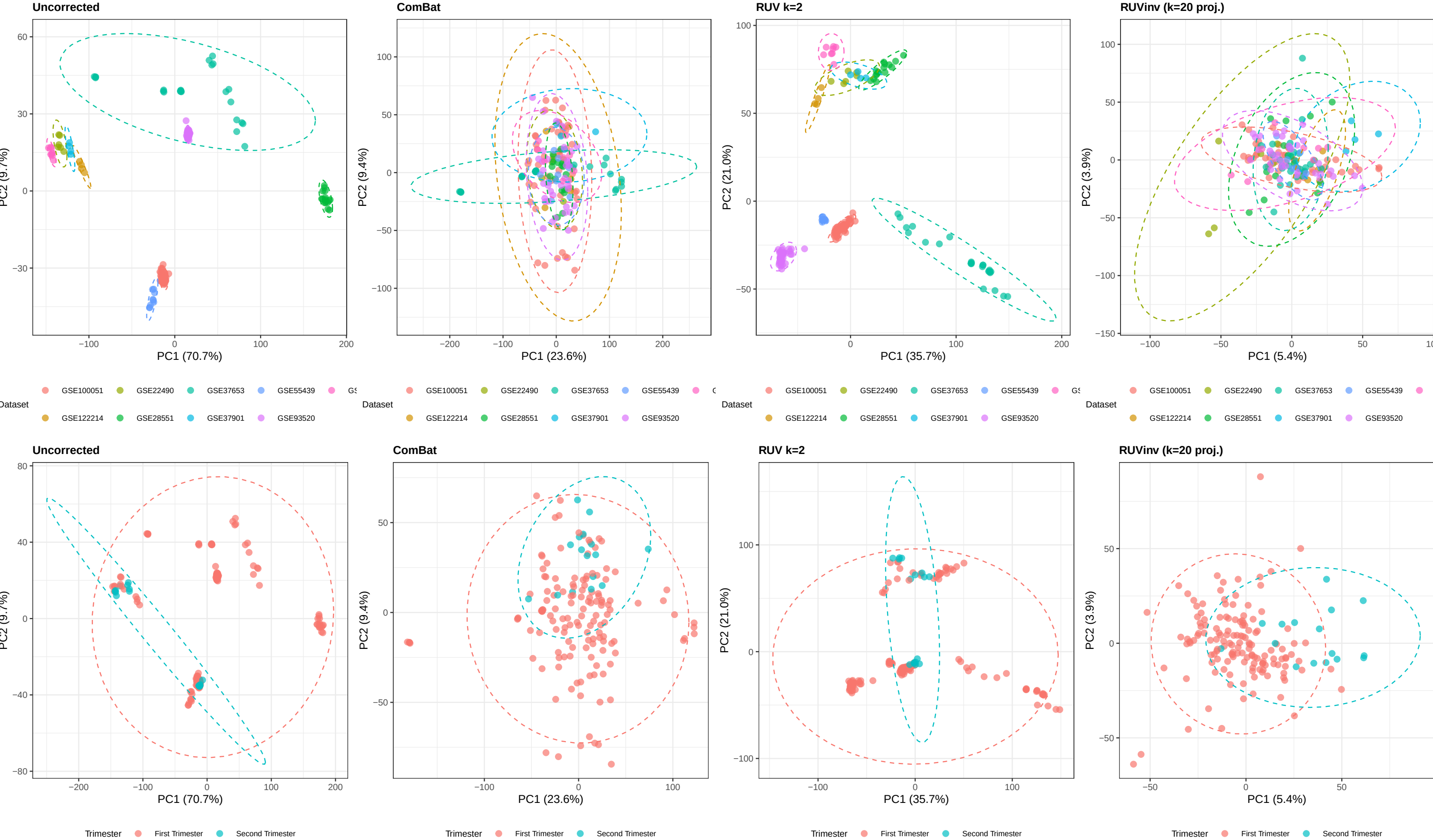
Top: colored by dataset (batch). Bottom: colored by trimester (biology).





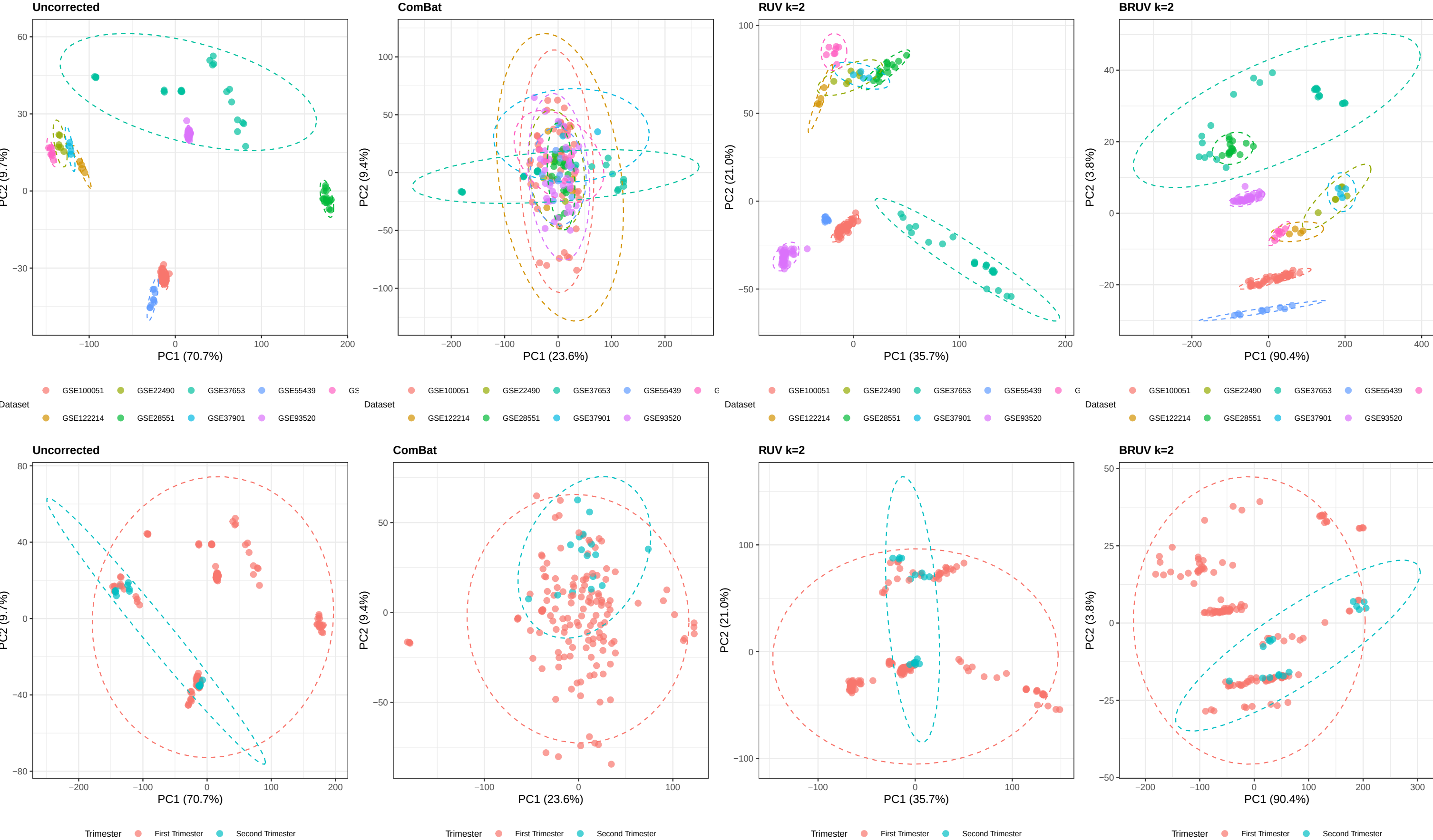
1st vs 2nd Trim – All Datasets – PCA: Uncorrected / ComBat / RUV k=2 / RUVinv

Top: dataset. Bottom: trimester. RUVinv visualized by projecting out top-20 SVD components.



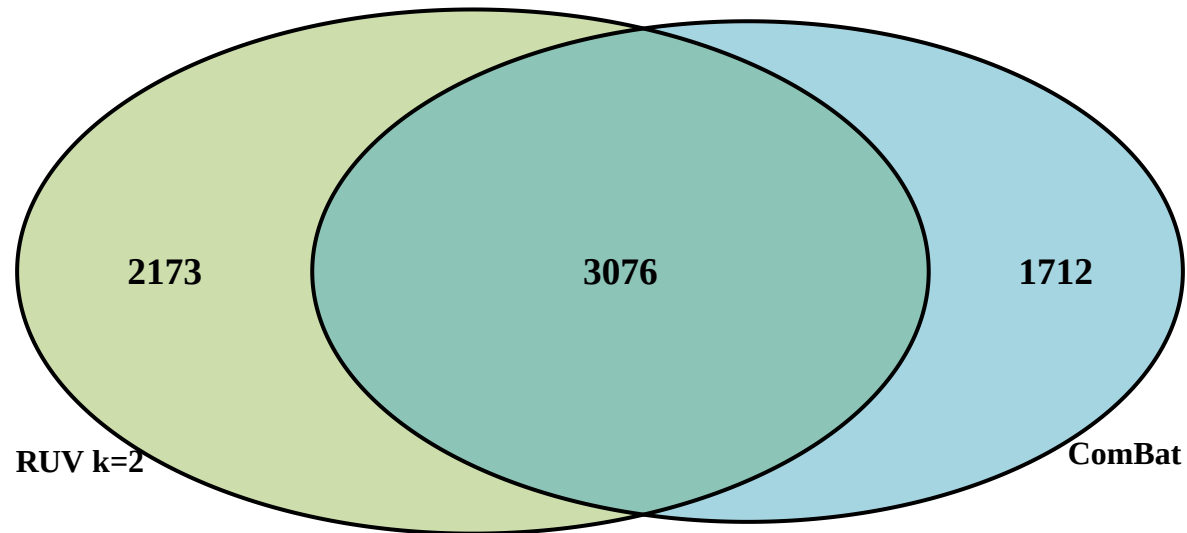
1st vs 2nd Trim – All Datasets – PCA: Uncorrected / ComBat / RUV k=2 / BRUV

Top: dataset. Bottom: trimester. BRUV estimates W from balanced datasets only, projects onto all.

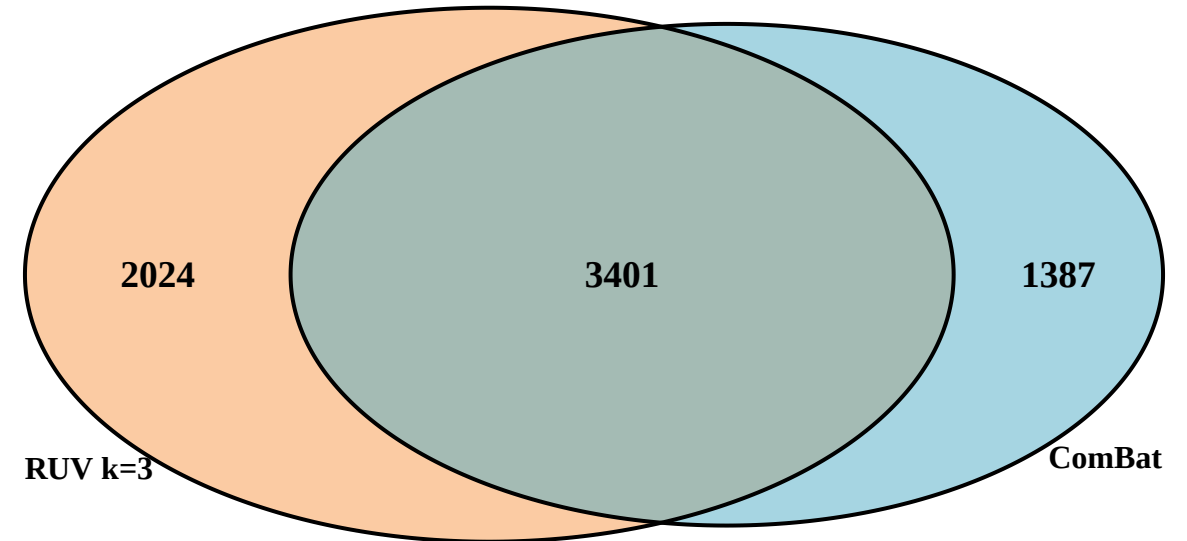


## 2nd Trim vs Term – All Datasets – DEG Venn Diagrams (FDR < 0.05)

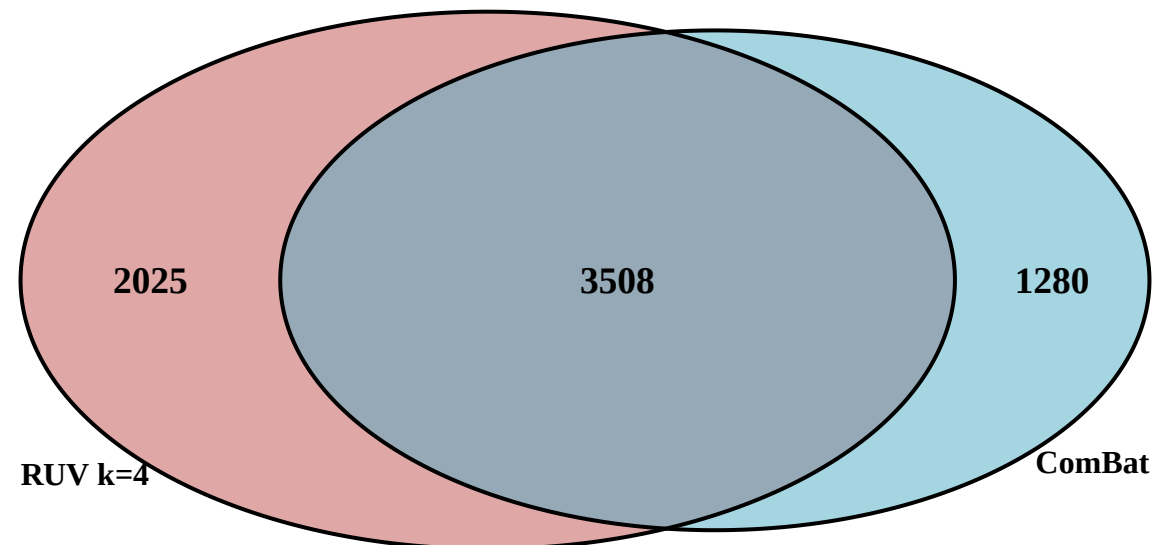
ComBat vs RUV k=2



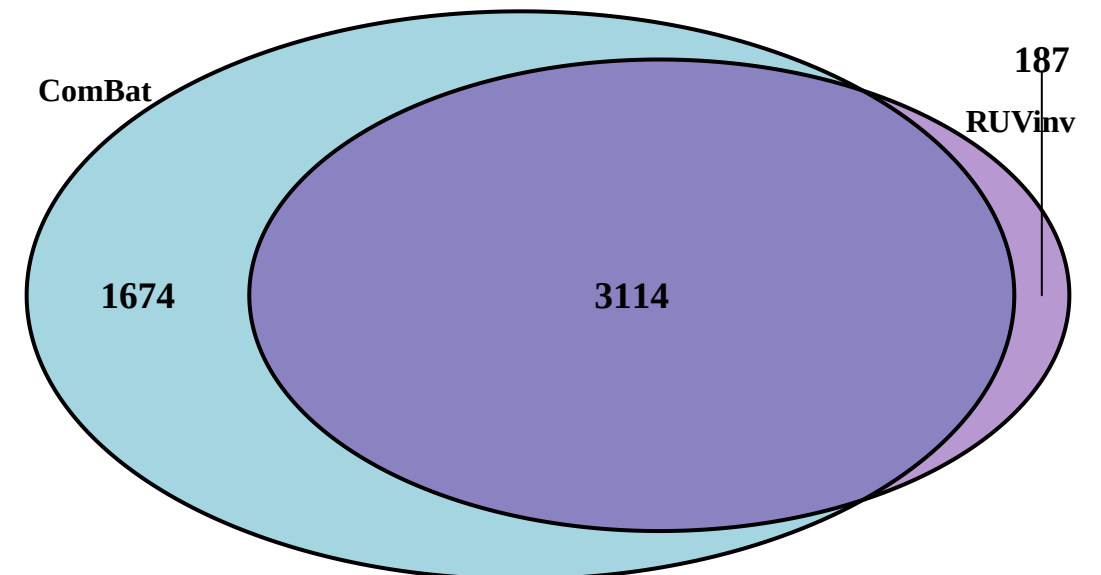
ComBat vs RUV k=3



ComBat vs RUV k=4

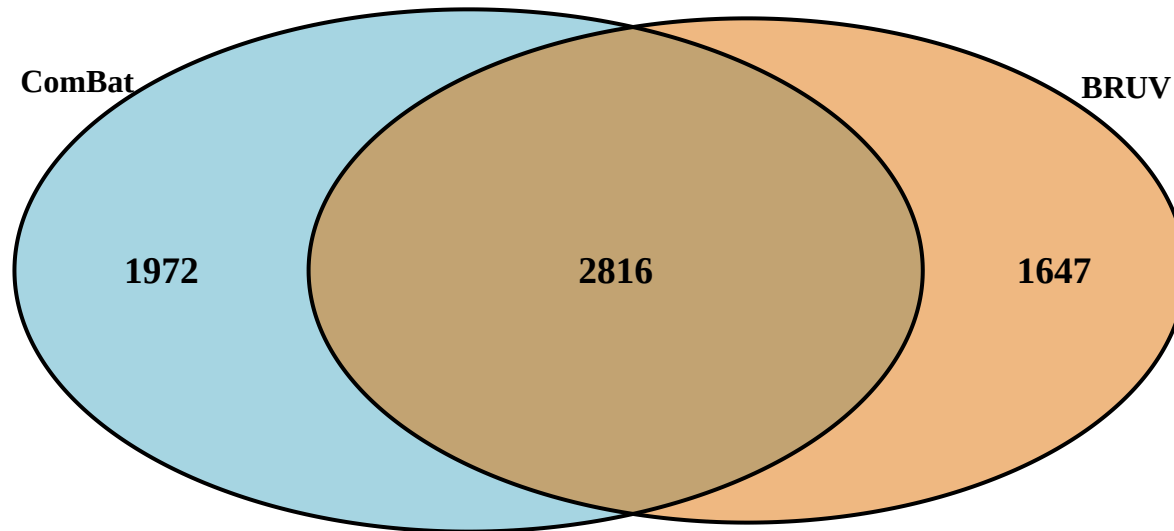


ComBat vs RUVinv



## 2nd Trim vs Term – All Datasets – BRUV Venn Diagrams (FDR < 0.05)

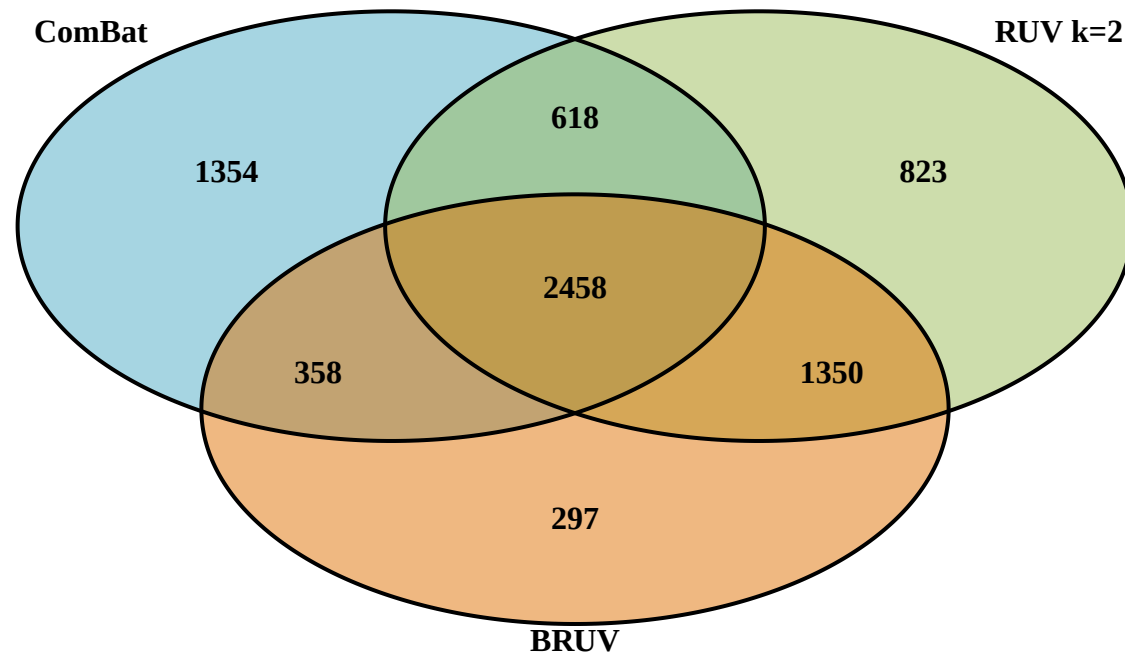
ComBat vs BRUV



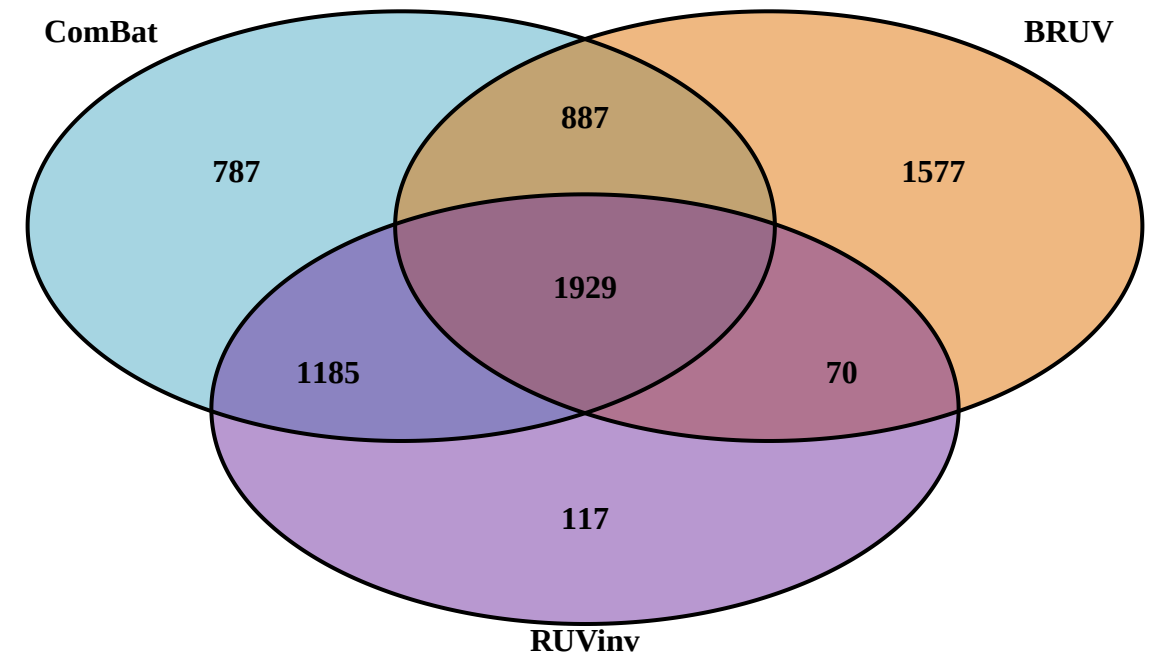
RUV k=2 vs BRUV



ComBat vs RUV k=2 vs BRUV

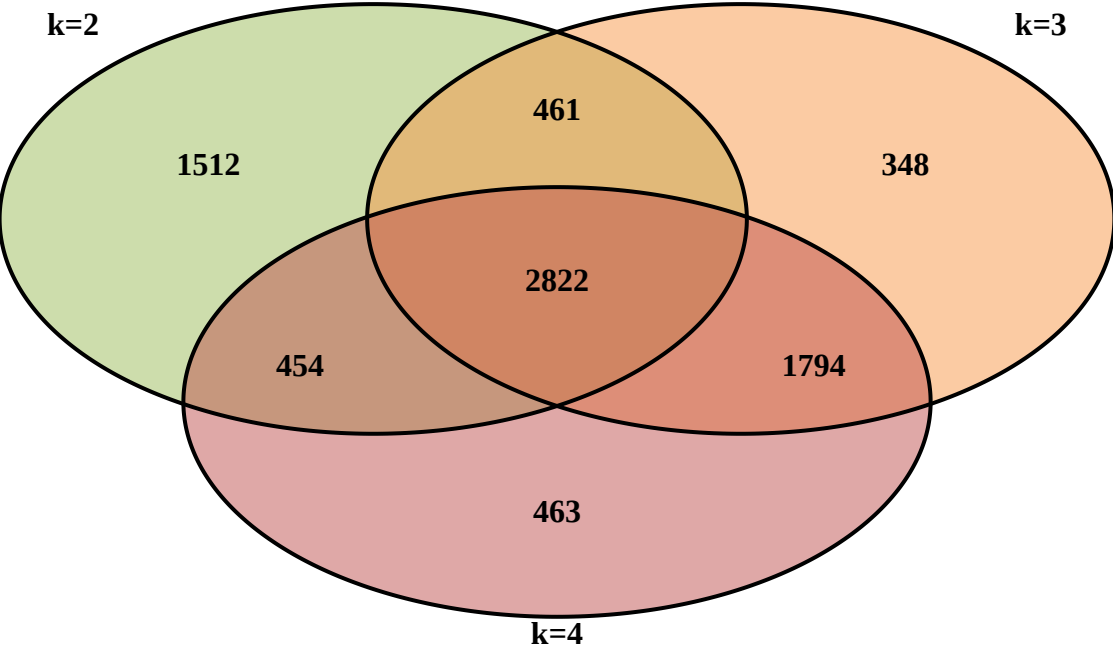


ComBat vs BRUV vs RUVinv

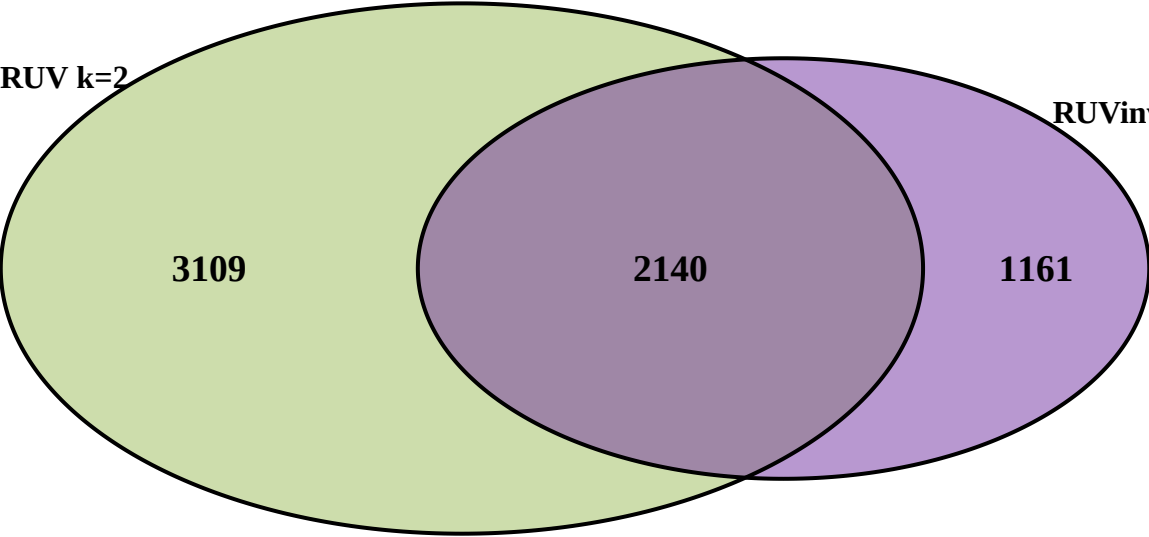


2nd Trim vs Term – All Datasets – RUV Variant Comparison (FDR < 0.05)

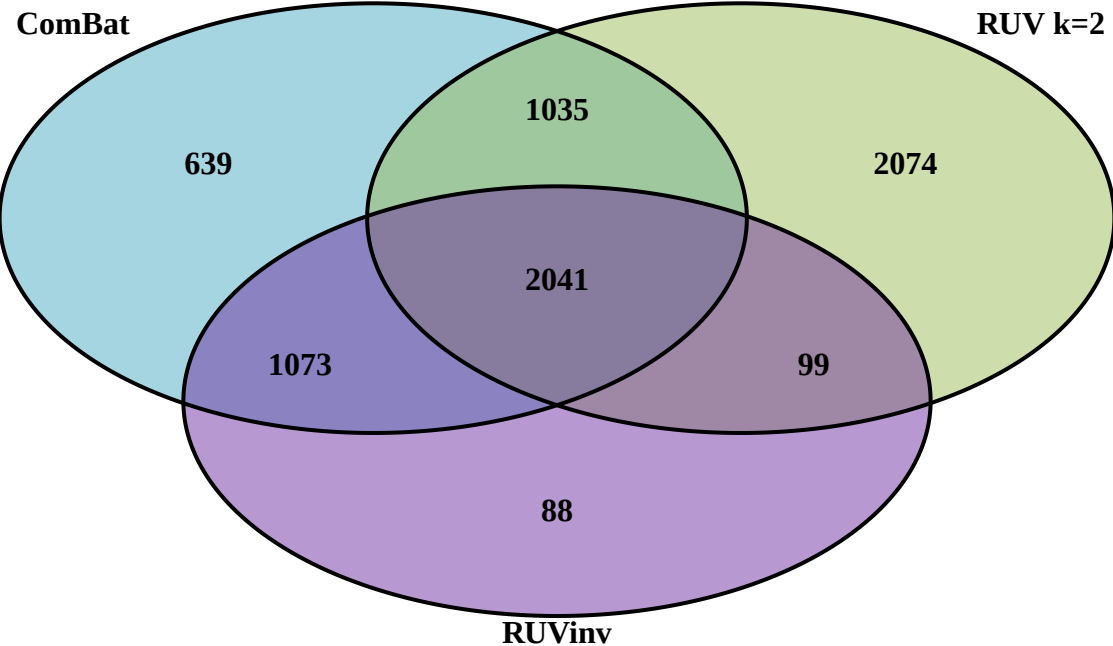
RUV k=2 vs k=3 vs k=4



RUV k=2 vs RUVinv

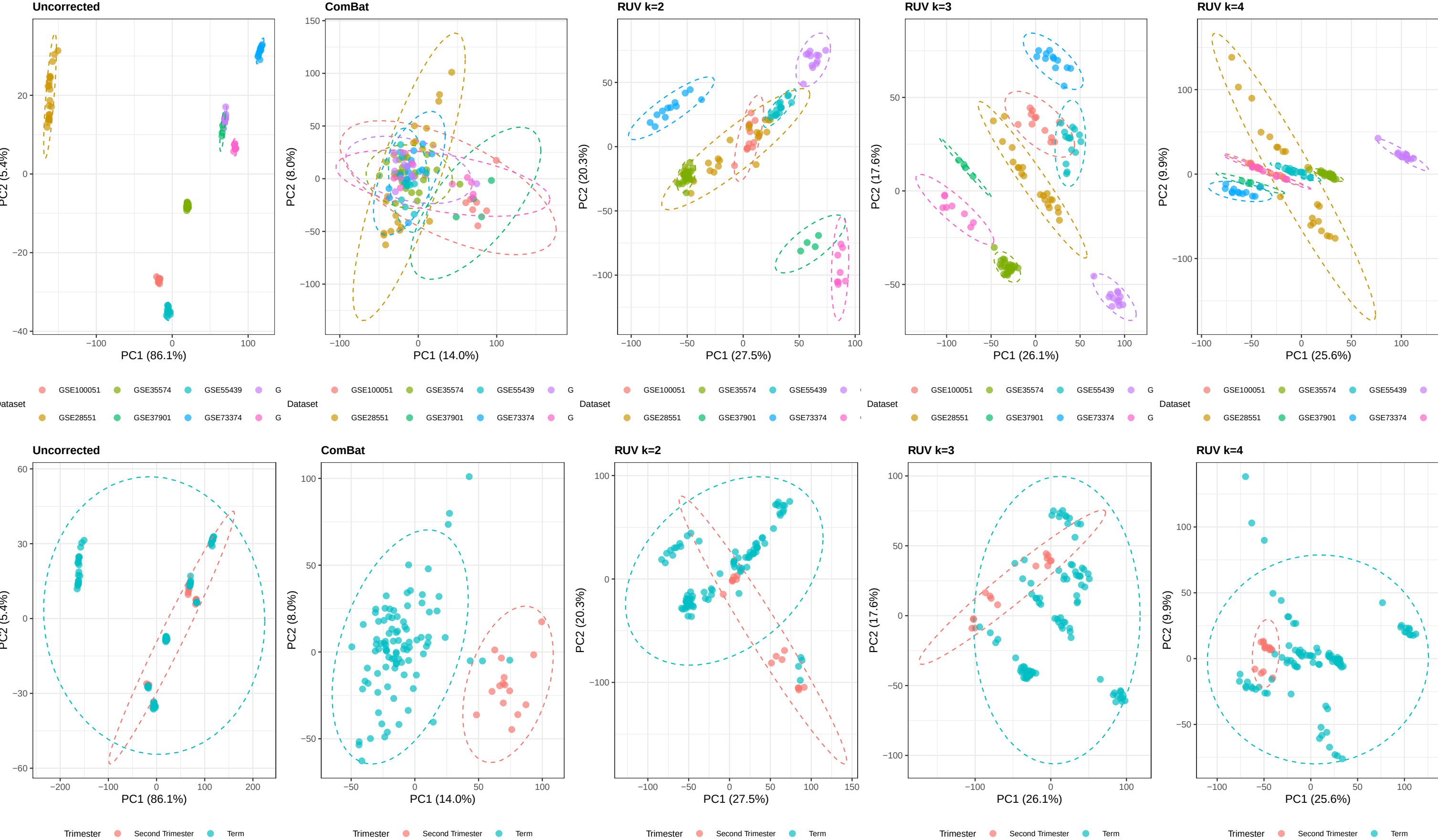


ComBat vs RUV k=2 vs RUVinv



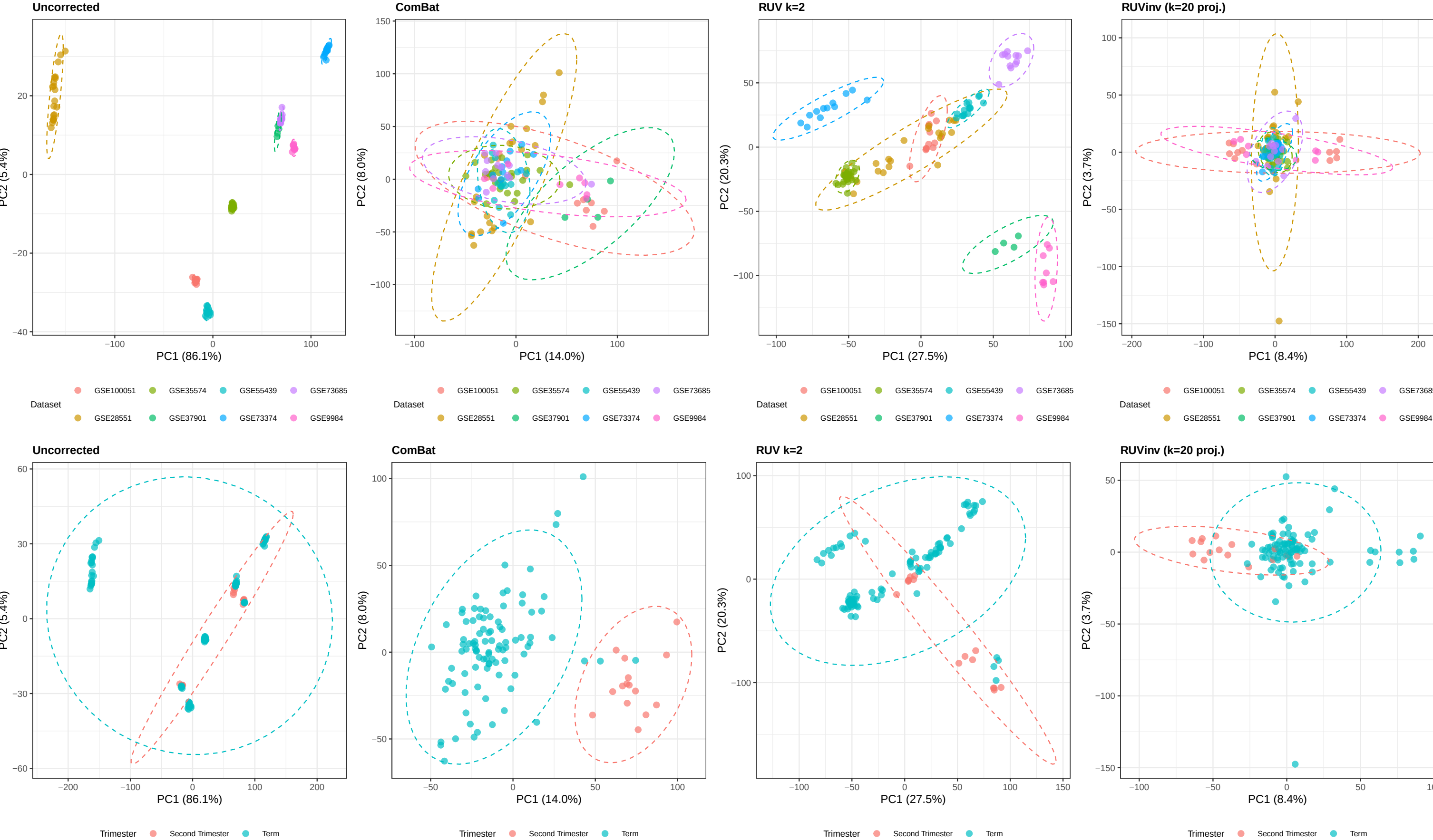
2nd Trim vs Term – All Datasets – PCA (k=2,3,4)

Top: colored by dataset (batch). Bottom: colored by trimester (biology).



2nd Trim vs Term – All Datasets – PCA: Uncorrected / ComBat / RUV k=2 / RUVinv

Top: dataset. Bottom: trimester. RUVinv visualized by projecting out top-20 SVD components.





2nd Trim vs Term – All Datasets – PCA: Uncorrected / ComBat / RUV k=2 / BRUV

Top: dataset. Bottom: trimester. BRUV estimates W from balanced datasets only, projects onto all.

